

CONCEPT OF RC CIRCUIT

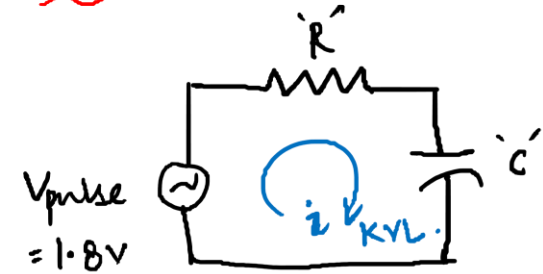
Er. Sailesh Mishra

B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI

DERIVATION OF CHARGING & DISCHARGING EQUATIONS OF A CAPACITOR

Consider a R-C ckt; with Resistance 'R' and Capacitance 'C'.



Putting KVL: We get

$$V_{\text{pulse}} - V_R - V_C = 0$$

$$V_{\text{pulse}} = V_R + V_C$$

$$V_{\text{pulse}} = i(t) \cdot R + \frac{q(t)}{C}$$

$$V_{\text{pulse}} = \frac{dq}{dt} \cdot R + \frac{q(t)}{C}$$

Comparing it with its General form; We get

$$\frac{dy}{dx} + P(x) \cdot y = Q(x)$$

$$\left\{ \frac{dq}{dt} + \frac{q(t)}{R \cdot C} = \frac{V_{\text{pulse}}}{R} \right\}$$

①
 (Integrating Factor)

$$\left. \begin{array}{l} x \rightarrow t \\ y \rightarrow q \end{array} \right\} \begin{array}{l} P(x) = \frac{1}{R \cdot C} \\ Q(x) = \frac{V_{\text{pulse}}}{R} \end{array}$$

$$\text{I.F.} \rightarrow e^{\int P(x) \cdot dx} = e^{t/R \cdot C}$$

We get;

$$\text{I} \rightarrow \frac{d}{dt} \left[q \cdot e^{t/R \cdot C} \right]$$

$$\text{II} \rightarrow \frac{V_{\text{pulse}}}{R} \cdot e^{t/R \cdot C}$$

Now; Dividing 'R' throughout the eqn; We get-

$$\frac{V_{\text{pulse}}}{R} = R \cdot \frac{dq}{dt} \left(\frac{1}{R} \right) + \frac{q(t)}{C} \left(\frac{1}{R} \right)$$

Now; Multiplying the Entire eqn By the Integrating Factor i.e. $\rightarrow$

$$e^{t/R \cdot C} \times \left\{ \frac{dq}{dt} + \frac{q(t)}{R \cdot C} \right\} = e^{t/R \cdot C} \times \frac{V_{\text{pulse}}}{R}$$

$$\Rightarrow \frac{V_{\text{pulse}}}{R} = \frac{dq}{dt} + \frac{q(t)}{R \cdot C} \rightarrow \text{highest order} = 1$$

Hence; it's a FIRST ORDER DIFFERENTIAL eqn !!

I

II

$$\Rightarrow q(t) \cdot e^{t/R \cdot C} = \int \frac{V_{\text{pulse}}}{R} \cdot e^{t/R \cdot C} dt = \underbrace{C \cdot V_{\text{pulse}} \cdot e^{t/R \cdot C}}_{\text{Capacitance}} + \underbrace{K}_{\text{Arbitrary Integration Constant} \dots}$$

$$\text{@ time 't' = 0 :- } \int \frac{V_{\text{pulse}}}{R} \cdot e^{t/R \cdot C} dt \Big|_0^0 = 0 = C \cdot V_{\text{pulse}} \cdot e^{t/R \cdot C} + K$$

$$= \boxed{C \cdot V_{\text{pulse}} + K} \Rightarrow K = -C \cdot V_{\text{pulse}}$$

$$\text{Similarly; @ time 't' = } C \cdot V_{\text{pulse}} (1 - e^{-t/R \cdot C}) = q(t).$$

{ $\because$ Since; $Q = CV$

$$\text{So; } V_C(t) = \frac{q(t)}{C} = \frac{q}{C} \cdot V_{\text{pulse}} (1 - e^{-t/R \cdot C}) \Rightarrow V_C(t) = V_{\text{pulse}} (1 - e^{-t/R \cdot C})$$

CHARGING

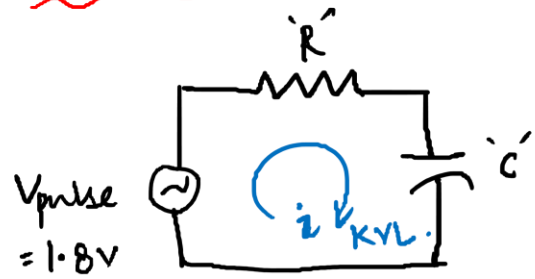
$$q(t) = C \cdot V(t)$$

$$\downarrow$$

$$V_{\text{pulse}} (1 - e^{-t/R \cdot C})$$

DISCHARGING EQUATION OF A CAPACITOR

Consider a R-C ckt; with Resistance 'R' and Capacitance 'C'.



Putting KVL: We get

$$V_R - V_C = 0$$

$$0 = V_R + V_C$$

$$0 = i(t) \cdot R + \frac{q(t)}{C}$$

$$0 = \frac{dq}{dt} \cdot R + \frac{q(t)}{C}$$

Now; dividing 'R' throughout the eqn; we get-

$$\frac{V_{\text{pulse}}}{R} = R \cdot \frac{dq}{dt} \left(\frac{1}{R} \right) + \frac{q(t)}{C} \left(\frac{1}{R} \right)$$

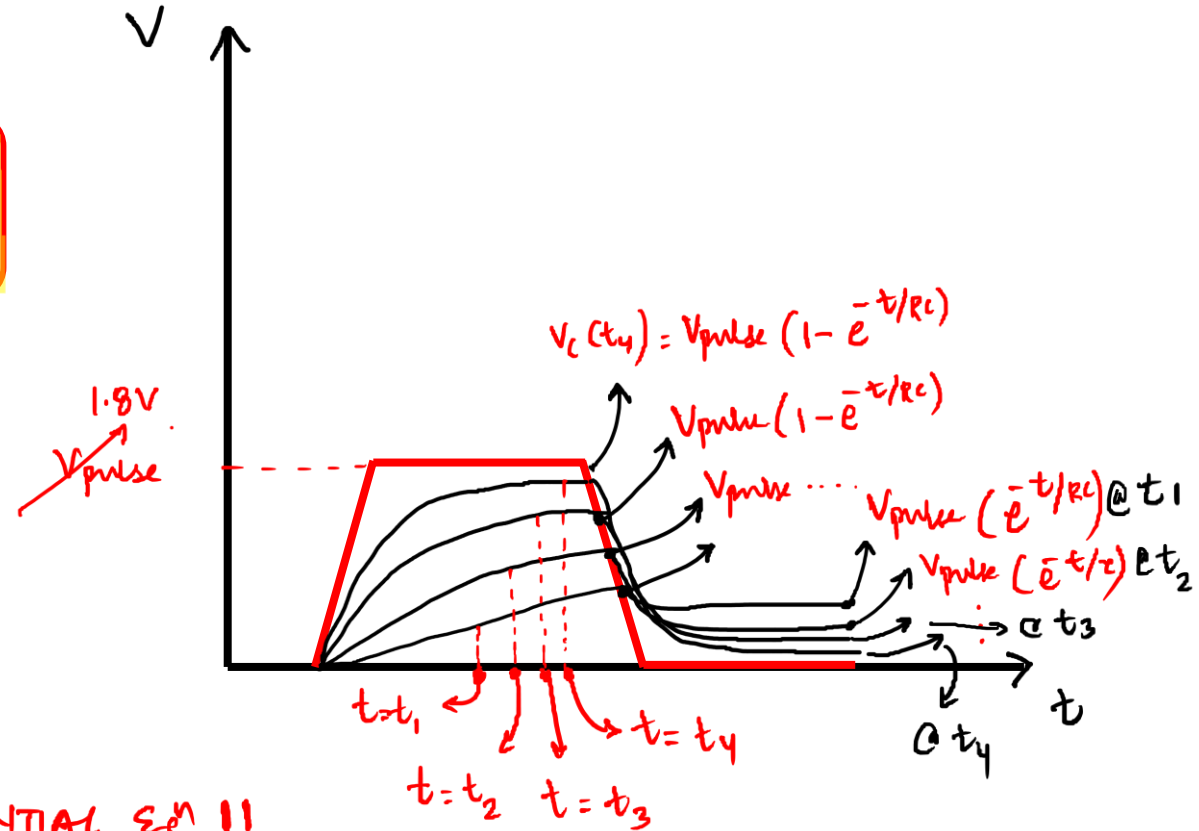
$$\Rightarrow 0 = \frac{dq}{dt} + \frac{q(t)}{R \cdot C} \rightarrow \text{highest order} = 1$$

Hence; it's a FIRST ORDER DIFFERENTIAL EQN !!

$$\Rightarrow \frac{dq}{dt} = -\frac{q}{R \cdot C} \Rightarrow q(t) = q_0 \cdot e^{-t/R \cdot C}$$

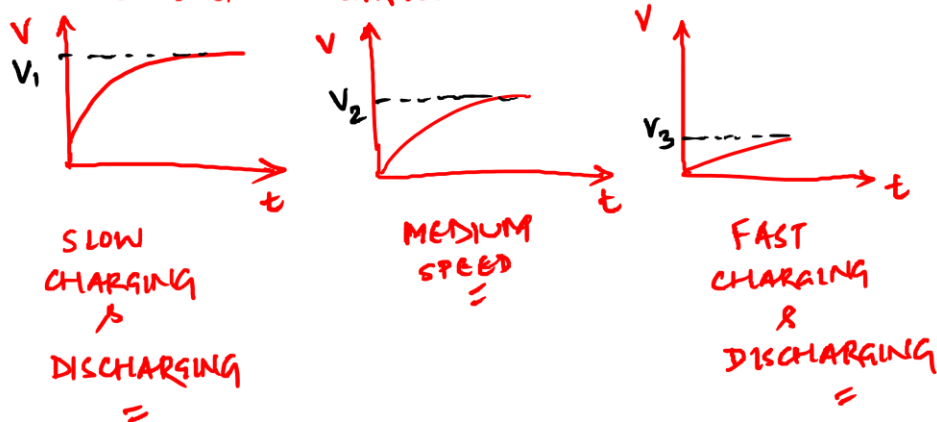
Similarly;
=

$$V_C(t) \text{ DISCHARGING} = V_{\text{pulse}} \times e^{-t/R \cdot C}$$



What is 1 τ , 5 τ & 10 τ

τ $\rightarrow$ are the Measures of HOW QUICKLY; A capacitor CHARGES and DISCHARGES...
[t_{ao}]

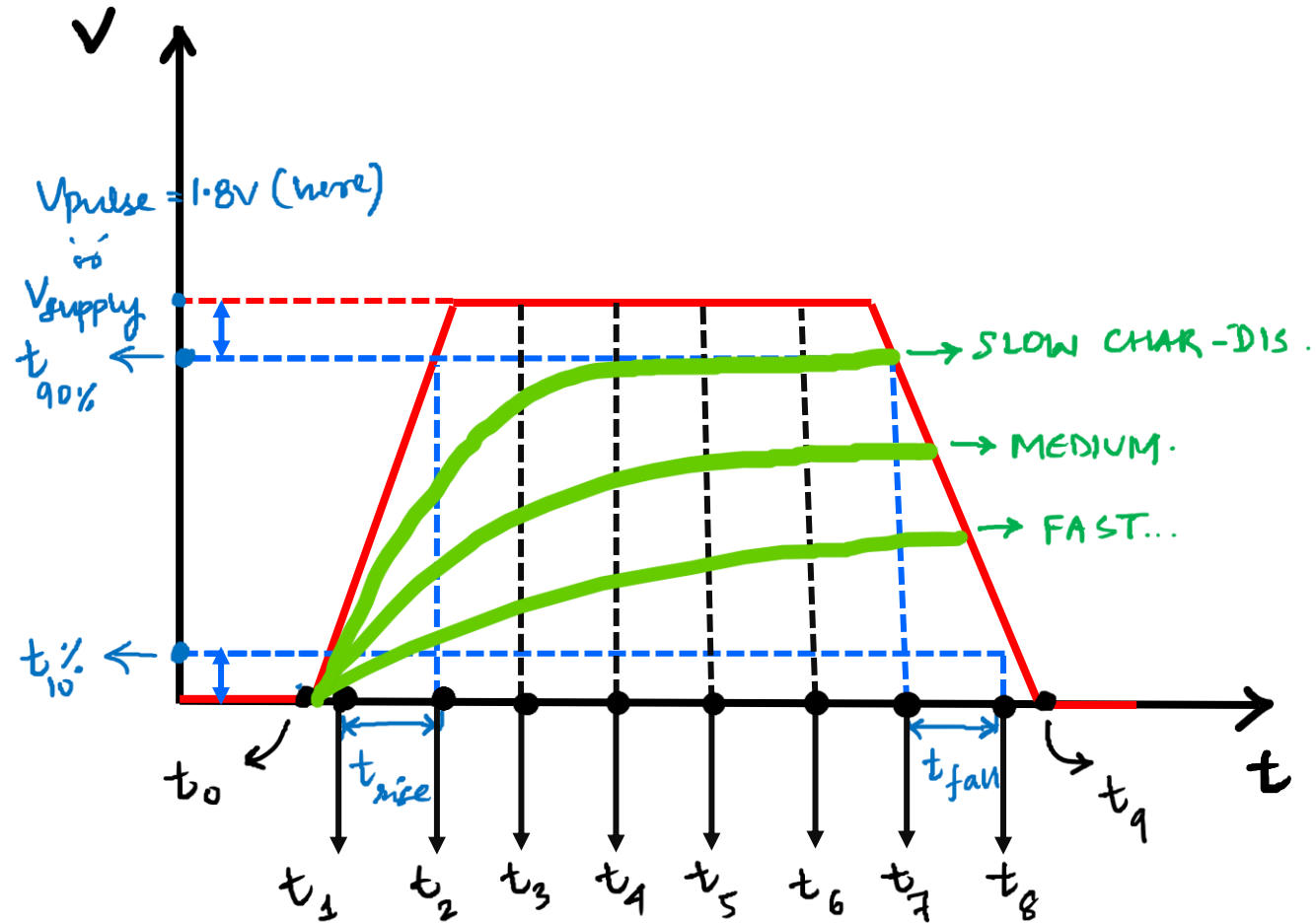


Considering 1.8V is the supply voltage.
 NOTE # :- More the speed of CHARGING & DISCHARGING, Less the Voltage $V_c(t)$.

$$\boxed{V_1 > V_2 > V_3}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $1.7999V \quad 1.15V \quad 0.6V$
 $= \quad = \quad =$

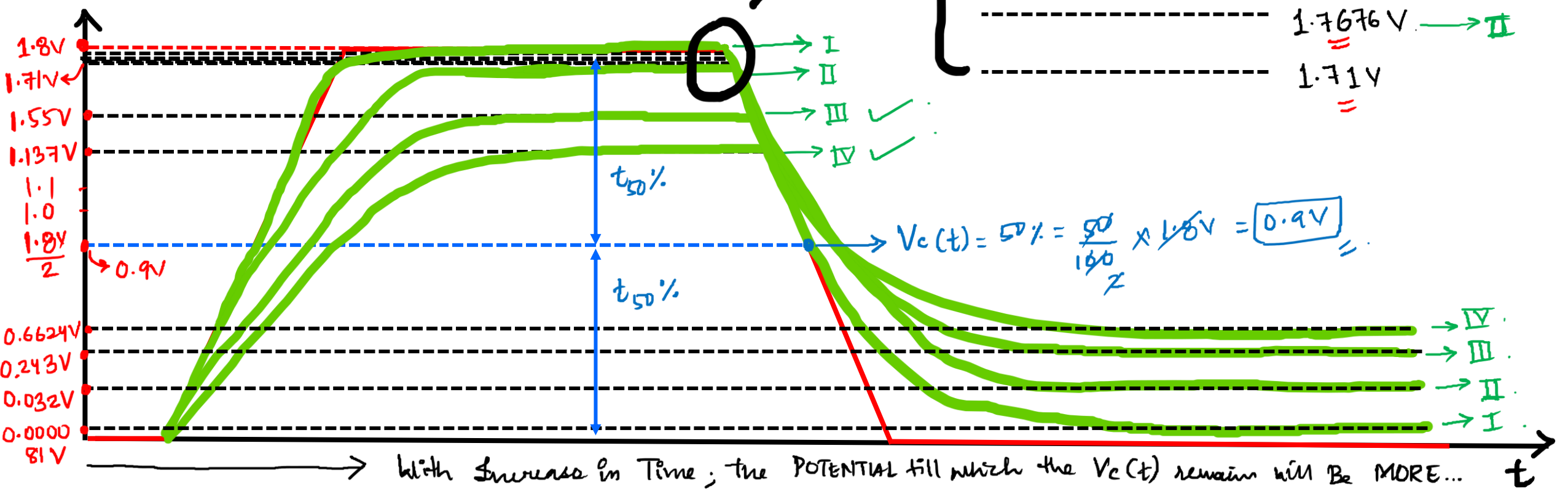
Remember :- A CAPACITOR NEVER FULLY CHARGES TO V_{dd} or FULLY DISCHARGES TO GND .



Charging and Discharging Percentage vs. Time Constants (τ)

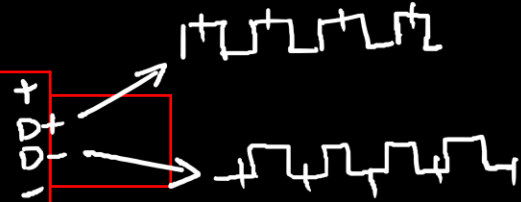
Time (t)	Charging $V(t) = V_f(1 - e^{-t/\tau})$	% Charged	Discharging $V(t) = V_i e^{-t/\tau}$	% Discharged
1τ	$V_f(1 - e^{-1}) = 0.632 V_f$	63.2%	$V_i e^{-1} = 0.368 V_i$	36.8%
2τ	$V_f(1 - e^{-2}) = 0.865 V_f$	86.5%	$V_i e^{-2} = 0.135 V_i$	13.5%
3τ	$V_f(1 - e^{-3}) = 0.950 V_f$	95.0%	$V_i e^{-3} = 0.0498 V_i$	4.98%
4τ	$V_f(1 - e^{-4}) = 0.982 V_f$	98.2%	$V_i e^{-4} = 0.0183 V_i$	1.83%
5τ	$V_f(1 - e^{-5}) = 0.993 V_f$	99.3%	$V_i e^{-5} = 0.0067 V_i$	0.67%
6τ	$V_f(1 - e^{-6}) = 0.9975 V_f$	99.75%	$V_i e^{-6} = 0.0025 V_i$	0.25%
7τ	$V_f(1 - e^{-7}) = 0.9991 V_f$	99.91%	$V_i e^{-7} = 0.00091 V_i$	0.091%
8τ	$V_f(1 - e^{-8}) = 0.99966 V_f$	99.966%	$V_i e^{-8} = 0.00034 V_i$	0.034%
9τ	$V_f(1 - e^{-9}) = 0.99988 V_f$	99.988%	$V_i e^{-9} = 0.00012 V_i$	0.012%
10τ	$V_f(1 - e^{-10}) = 0.99995 V_f$	99.995%	$V_i e^{-10} = 0.000045 V_i$	0.0045%

PRACTICAL TOPOLOGY & OUTPUT WAVEFORM ANALYSIS

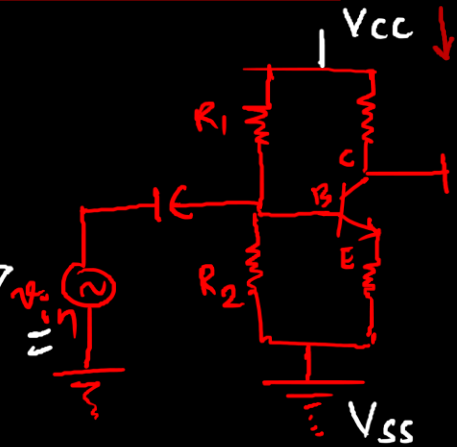


USB
2.0
3.0

Pendrive/dongle

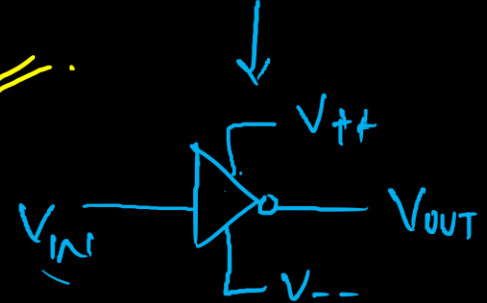
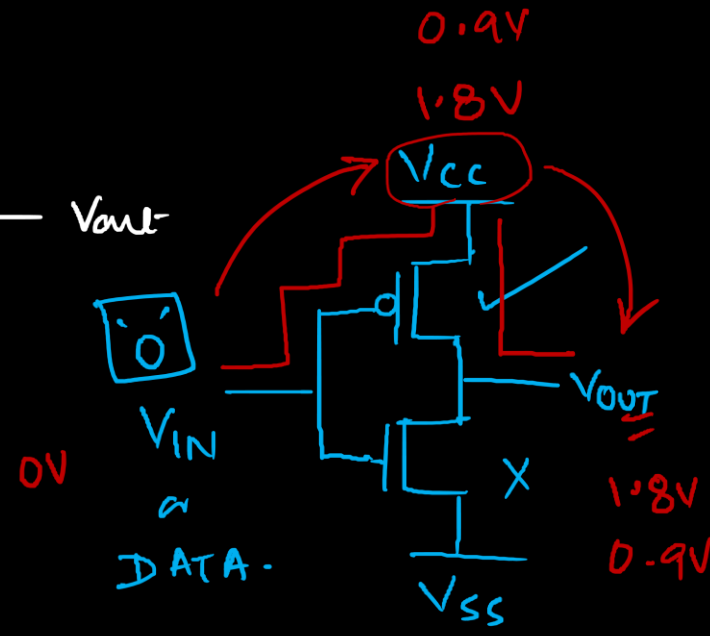
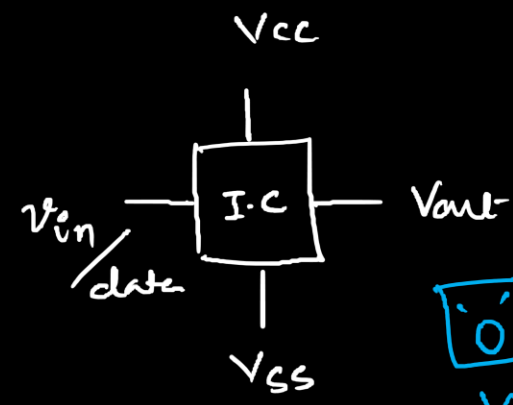
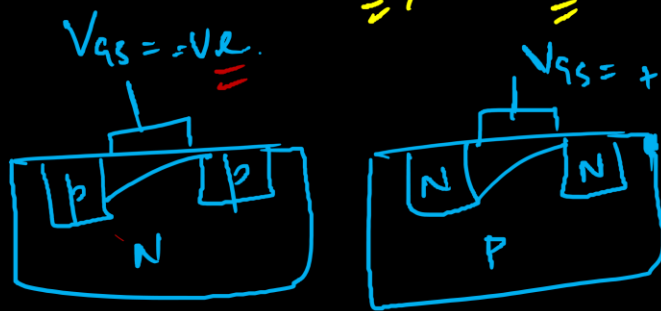
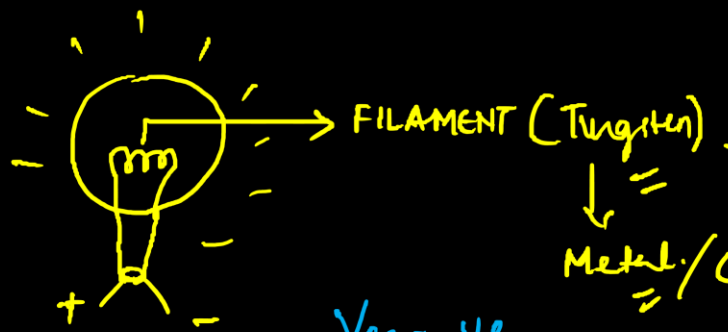
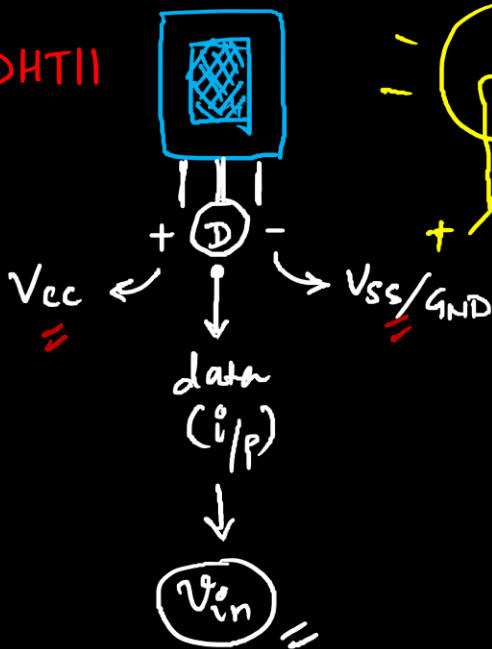


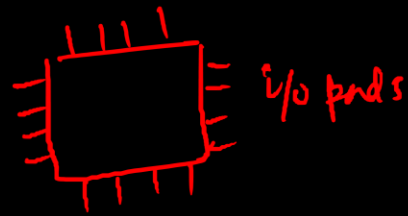
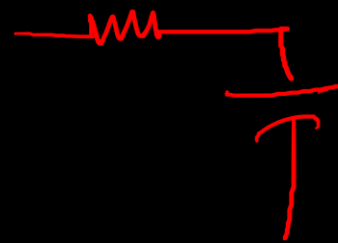
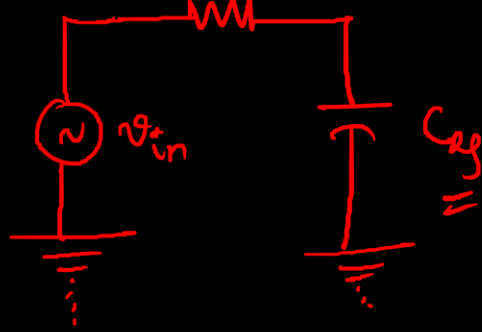
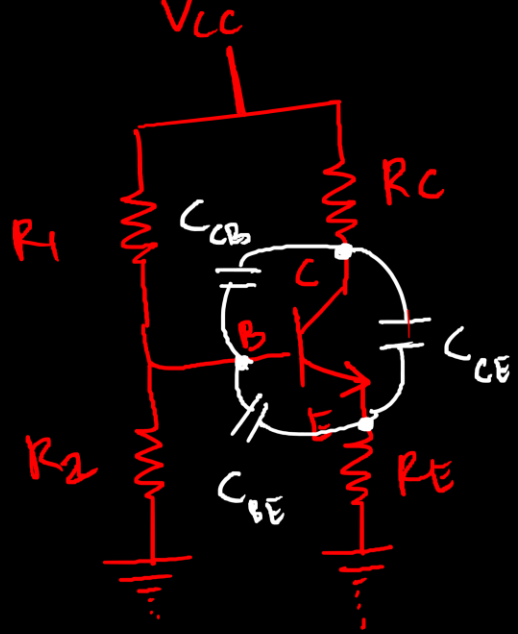
Signal Encoding



I.C $\rightarrow$ R, C, D, M, BJT

DHT11

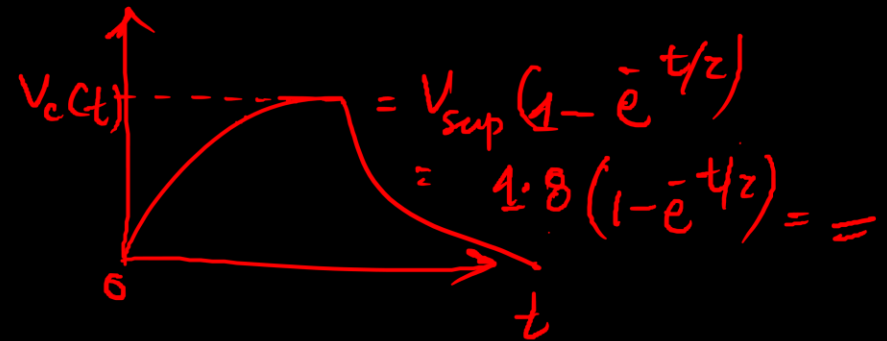


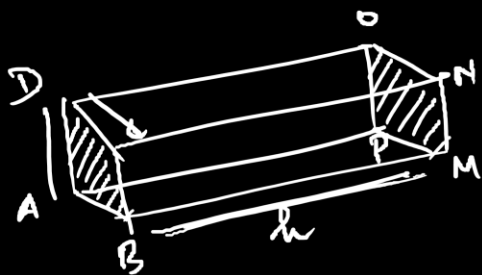


$$R_{eq} = R_{th} \parallel (R_C \parallel R_E) = \left(\frac{R_1 R_2}{R_1 + R_2} \right) \parallel \left(\frac{R_C \cdot R_E}{R_C + R_E} \right) = \frac{A \cdot B}{A + B} = R$$

ELMoore Delay Model.

$$Z = R \cdot C$$





• MODEL file:

ext

• pdf

• pptx

• odt / • docx

• csv / • xlxx



CMOS Inverter
= =

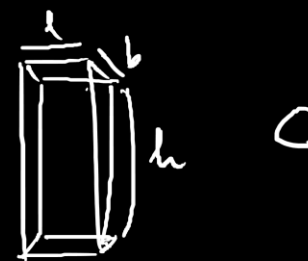
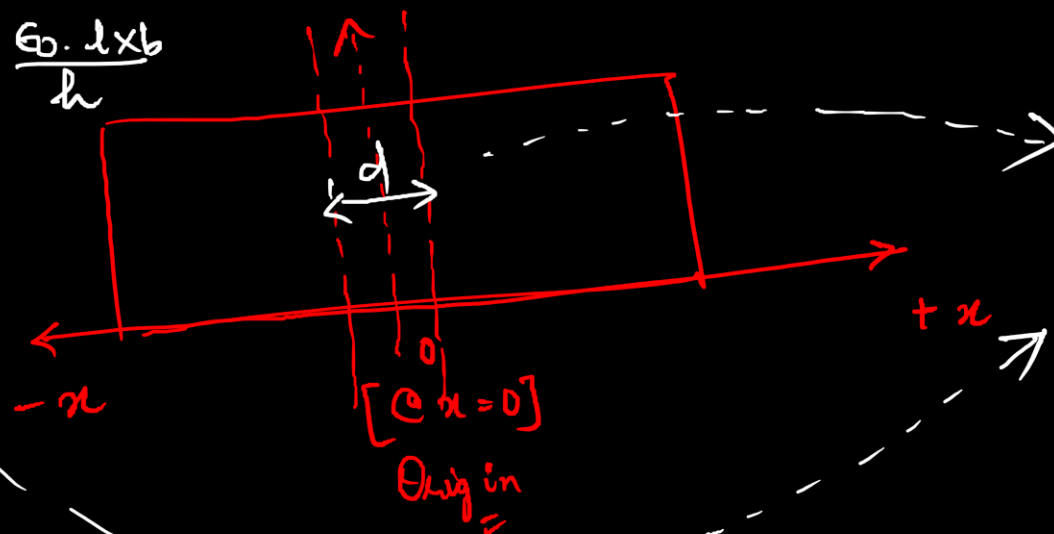
Cadence

Model file

$$C = \epsilon_0 \cdot \frac{A}{d}$$

$$= \epsilon_0$$

$$\square ABCD \rightarrow \text{cap} \rightarrow C_{ABCD} = \frac{\epsilon_0 \cdot A}{d} = \frac{\epsilon_0 \cdot l \times b}{h}$$



$$C_{j0} = \frac{\epsilon_0 \cdot A}{d} \rightarrow \frac{l \times b}{h}$$

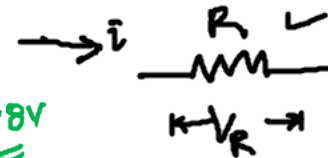
$$= 1 \text{ pF}$$

BASICS ...

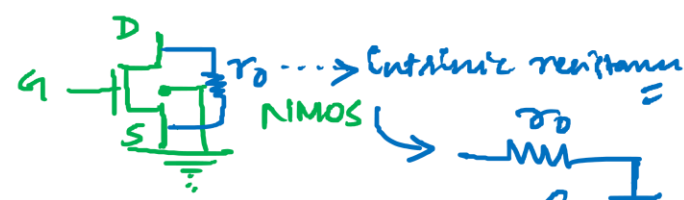
$$V_C(t) \text{ charging} = V_{\text{Supply}} \times (1 - e^{-t/\tau})$$

$$V_C(t) = 1.8 \times (1 - e^{-t/\tau}) \rightarrow R \cdot C$$

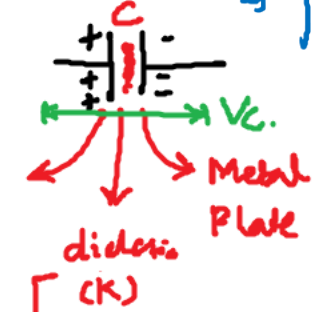
should always be nearer to V_{Supply} . i.e. 1.8V



$$i = \frac{V}{R}$$



Similarly;



ex. SiO_2 : VL61
K = 3.97 (4)

Al_2O_3 = 9.0 (v. high)

Air = 1.0 (K).

H_2O_2 = 20-25.

$q = ne + ne$

Voltage (in volts).

capacitance = $10 \mu\text{F}$
 $= 10 \times 10^{-6} \text{ F}$
 $V = 5 \text{ V}$
 $Q = C \cdot V$
 $= 10 \mu\text{F} \times 5$
 $= 50 \mu\text{C}$

$V = \frac{Q}{C}$
 $Q = CV$

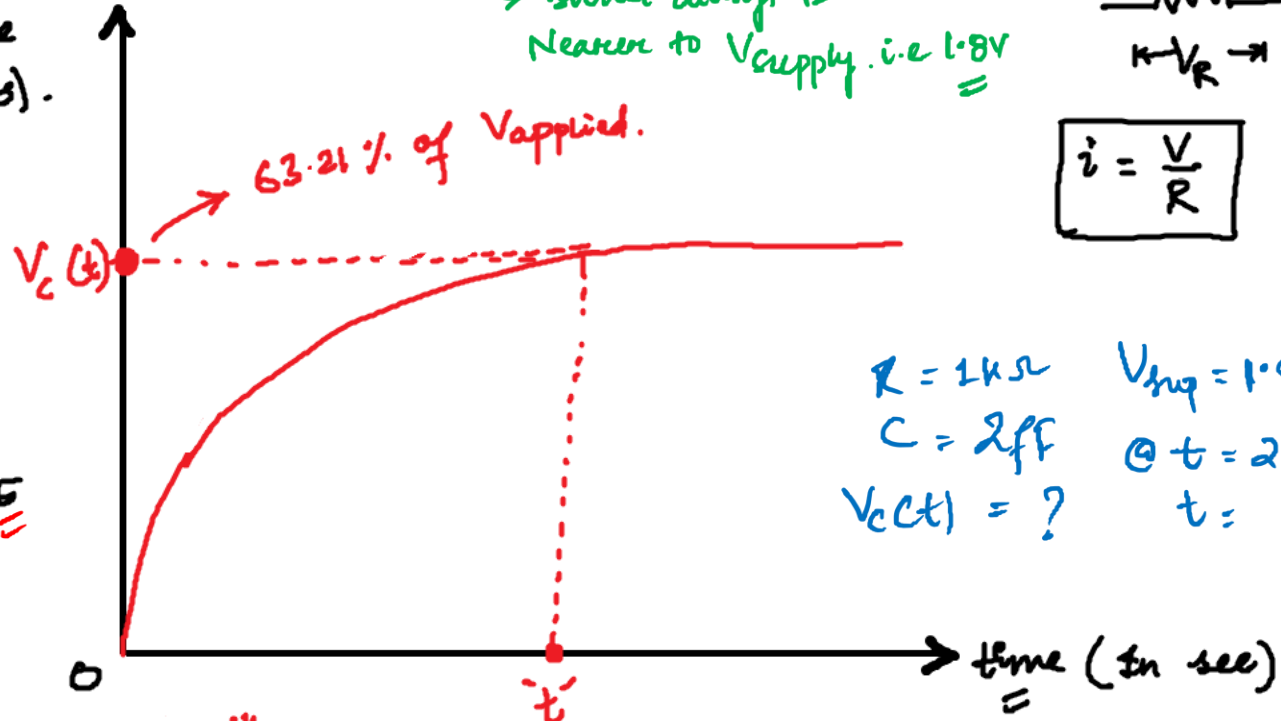
$Q = ne$

$50 \mu\text{C} = ne = 3.125 \times 10^{14}$
 1.6×10^{-19}
 $= 300 \text{ no. of } e^-$

15D +ve → +ve METAL PLATE.
 15D -ve → -ve METAL PLATE.



$Q = CV$
 V or Battery is always const.



$R = 1 \text{ k}\Omega$
 $C = 2 \text{ fF}$
 $V_C(t) = ?$
 $V_{\text{avg}} = 1.8$
 $@ t = 2 \tau$
 $t =$

V
(mV)
volts

V_{app}

1.8V

1.2V → 1.25V

0.9

0.45

0 → 0V

0 1 2 3 4 5

5ms

$i' \rightarrow 1.8V$

$V_C(t) : \text{CHARGING} \rightarrow V_C(t) = V_{app} (1 - e^{-t/\tau})$
 $1.25V = 1.8V (1 - e^{-t/\tau})$

$\tau = R \cdot C$
 $= 1k\Omega \times 10\mu F$
 $= 1 \times 10^3 \times 10 \times 10^{-6} F$

$V_C(t) : \text{DISCHARGING} \rightarrow V_C(t) = V_{app} \times (e^{-t/\tau})$

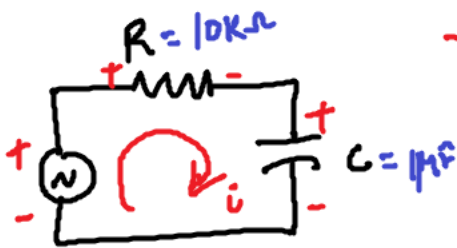
$1.8 (1 - e^{-\frac{3ms}{\tau}}) = 1.8 (1 - e^{-\frac{3 \times 10^{-3}s}{R \cdot C}})$
 $= 1.8 (1 - e^{-\frac{3 \times 10^{-3}}{10 \times 10^{-6}}}) = 1.8 (1 - e^{-\frac{3}{10}})$



Example-1 :-



$V_{\text{pulse}} = 1.8\text{V}$



$$\rightarrow V_{\text{pulse}} - V_R - V_C = 0$$

$$\Rightarrow 1.8 - V_R - V_C = 0$$

$\Rightarrow 1.8 - V_C = V_R$: voltage drop across $R = 10\text{k}\Omega = 10 \times 10^3 \Omega$.
voltage drop across $C = 1\mu\text{F} = 1 \times 10^{-6} \text{F}$.

$$P = \frac{V^2}{R} + \frac{1}{2} CV^2$$

Time period = 8ms
 $T_{\text{ON}} = 4\text{ms}$
 $T_{\text{rise}} = T_{\text{fall}} = 0\text{s}$

Q1 Find $i \rightarrow R$

Q2 Find $i \rightarrow C$

$$\tau = R \cdot C = 10\text{k}\Omega \times 1\mu\text{F} = 10 \times 10^3 \Omega \times 10^{-6} \text{F} = 10^{-2} \text{s}$$

$$\rightarrow i_R(t) = \frac{V}{R} = \frac{V_N}{R} = \frac{V_{\text{pulse}} - V_C}{R} = \frac{1.8\text{V} - V_{\text{app}} \times (1 - e^{-t/\tau})}{R}$$

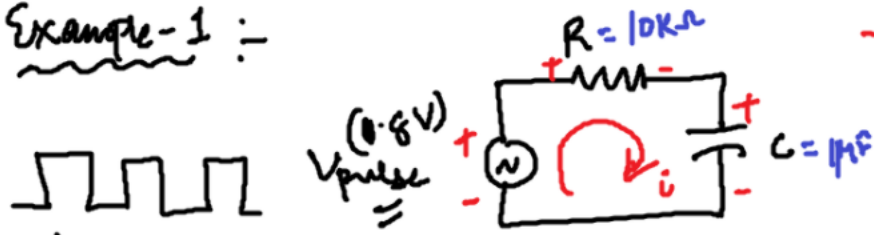
$$\rightarrow i_C(t) = C \cdot \frac{dV}{dt} = \frac{1\mu\text{F} \times V_{\text{app}} \times (1 - e^{-t/\tau})}{t}$$

meet.google.com is sharing your screen.

Stop sharing

Hide

Example-1 :



Time period = 8ms

$T_{ON} = 4ms$

$T_{rise} = T_{fall} = 0s$

Q1: Find $V_C(t)$ at $2ms$; when CAPACITOR is charging...

Q2: Find $V_C(t)$ at $5ms$; when CAPACITOR is discharging...

$$\rightarrow V_{pulse} - V_R - V_C = 0$$

$$\Rightarrow 1.8 - V_R - V_C = 0$$

$\Rightarrow 1.8 - V_C = V_R$ $\therefore$ voltage drop across $R = 10k\Omega = 10 \times 10^3 \Omega$.
voltage drop across $C = 1\mu F = 1 \times 10^{-6} F$.

$$\rightarrow \tau = R \cdot C = 10k\Omega \times 1\mu F = 10 \times 10^3 \Omega \times 10^{-6} F = 10^{-2} = 10ms$$

$$P = \frac{V^2}{R} + \frac{1}{2} CV^2$$

$$\rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau})$$

$$\rightarrow V_C(t) = V_{app} \times e^{-t/\tau}$$

$t: @ 0ms \rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-0/10 \times 10^{-3}}) = 1.8 \times (1 - 1) = 0V$ ✓

$t: @ 1ms \rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V (1 - e^{-\frac{1ms}{10ms}}) = 1.8 (1 - e^{-0.1}) = 0.171V$ ✓

$t: @ 2ms \rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-\frac{2ms}{10ms}}) = 1.8 (1 - e^{-0.2}) = 0.326V$ ✓

$$\uparrow P_{\text{avg}} = \frac{1}{2} \cdot C_L \cdot V_{DD}^2 \times f \uparrow$$


$$t: @ 0\text{ms} \rightarrow V_c(t) = V_{\text{app}} \times (1 - e^{-t/\tau}) = 1.8\text{V} \times (1 - e^{-0}) = 1.8 \times (1 - 1) = 0$$

$$t: @ 1\text{ms} \rightarrow V_c(t) = V_{\text{app}} \times (1 - e^{-t/\tau}) = 1.8\text{V} \times (1 - e^{-\frac{1\text{ms}}{10\text{ms}}}) = 1.8 (1 - e^{-0.1}) = \boxed{0.171\text{V}} \checkmark$$

$$t: @ 2\text{ms} \rightarrow V_c(t) = V_{\text{app}} \times (1 - e^{-t/\tau}) = 1.8\text{V} \times (1 - e^{-\frac{2\text{ms}}{10\text{ms}}}) = 1.8 (1 - e^{-0.2}) = \boxed{0.326\text{V}} \checkmark$$

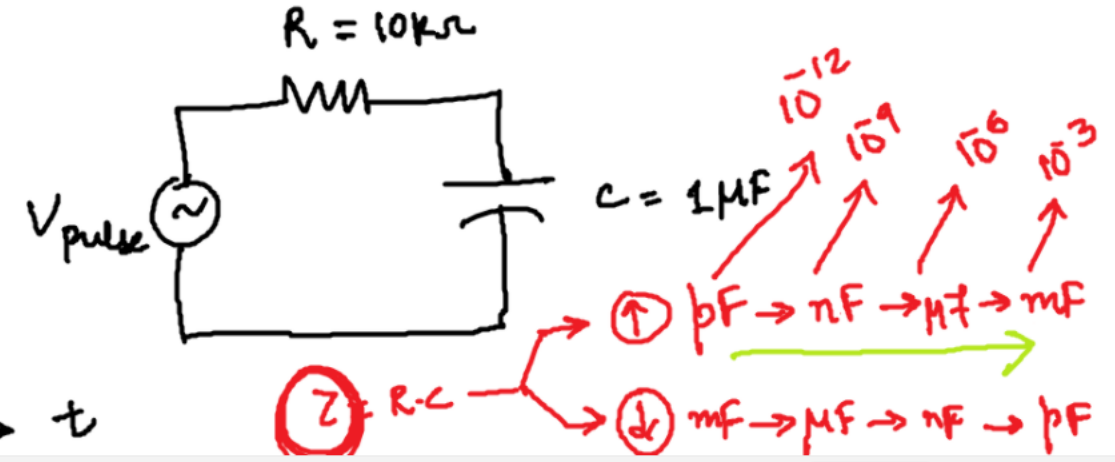
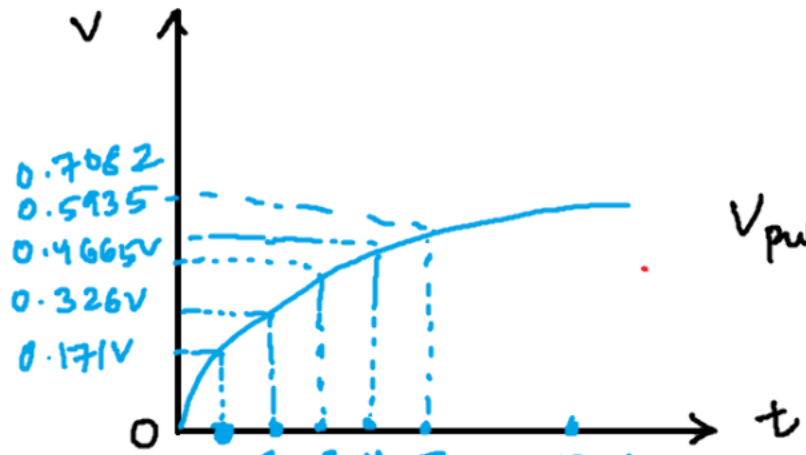
$$t: @ 3\text{ms} \rightarrow V_c(t) = V_{\text{app}} \times (1 - e^{-t/\tau}) = 1.8 (1 - e^{-0.3}) = \boxed{0.4665\text{V}} \checkmark$$

$$t: @ 4\text{ms} \rightarrow V_c(t) = 1.8 (1 - e^{-0.4}) = \boxed{0.5934\text{V}}$$

$$t: @ 5\text{ms} \rightarrow V_c(t) = 1.8 (1 - e^{-0.5}) = \boxed{0.7082\text{V}} \rightarrow 6\text{ms} = \boxed{0.8121\text{V}} \rightarrow 7\text{ms} = \boxed{0.9041\text{V}} \rightarrow 8\text{ms} = \boxed{0.9912\text{V}}$$

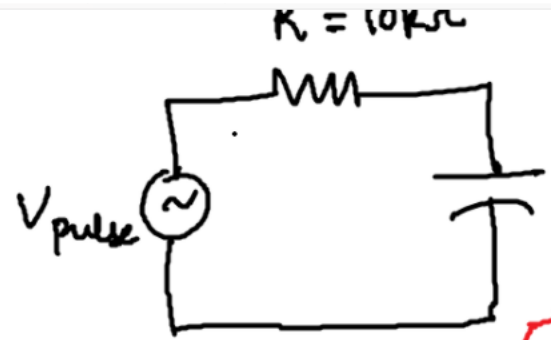
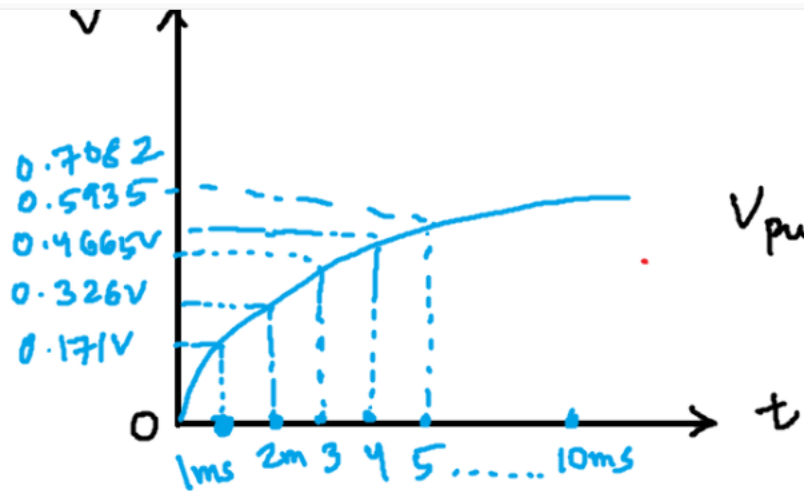
$\uparrow$ → FASTER charging
discharging

$\downarrow$ → SLOWER charging
discharging



$Z \uparrow \rightarrow$ FASTER charging discharging

$Z \downarrow \rightarrow$ SLOWER charging discharging



$C = 1\mu F$

$\uparrow$ pF $\rightarrow$ nF $\rightarrow$ μF $\rightarrow$ mF

$\downarrow$ mF $\rightarrow$ μF $\rightarrow$ nF $\rightarrow$ pF

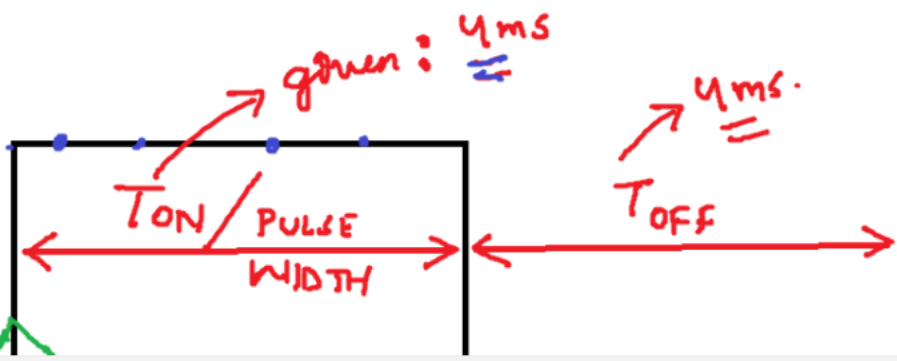
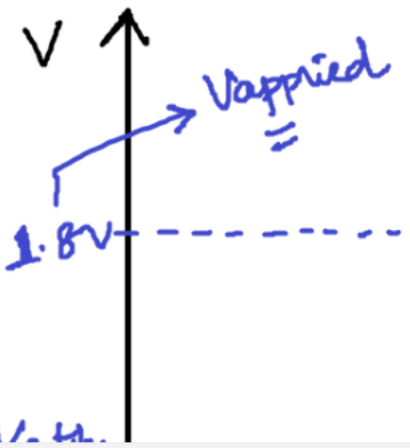
(10^3) (10^{12})

10^{-12} 10^{-9} 10^{-6} 10^{-3}

$$V_C(t) = V_{pulse} \times (1 - e^{-t/\tau})$$

$$= 1.8V \left(1 - e^{-\frac{1\mu s}{10\mu s}}\right) = 1.8 \times (1 - e^{-0.1})$$

$$= \boxed{0.171V}$$



@ 2ms (t)

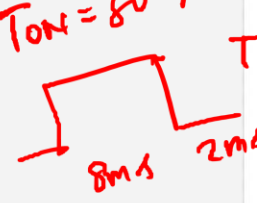
$V(t) = 1.8 \times (1 - e^{-0.2})$

50% DUTY CYCLE $T = 10\text{ms}$

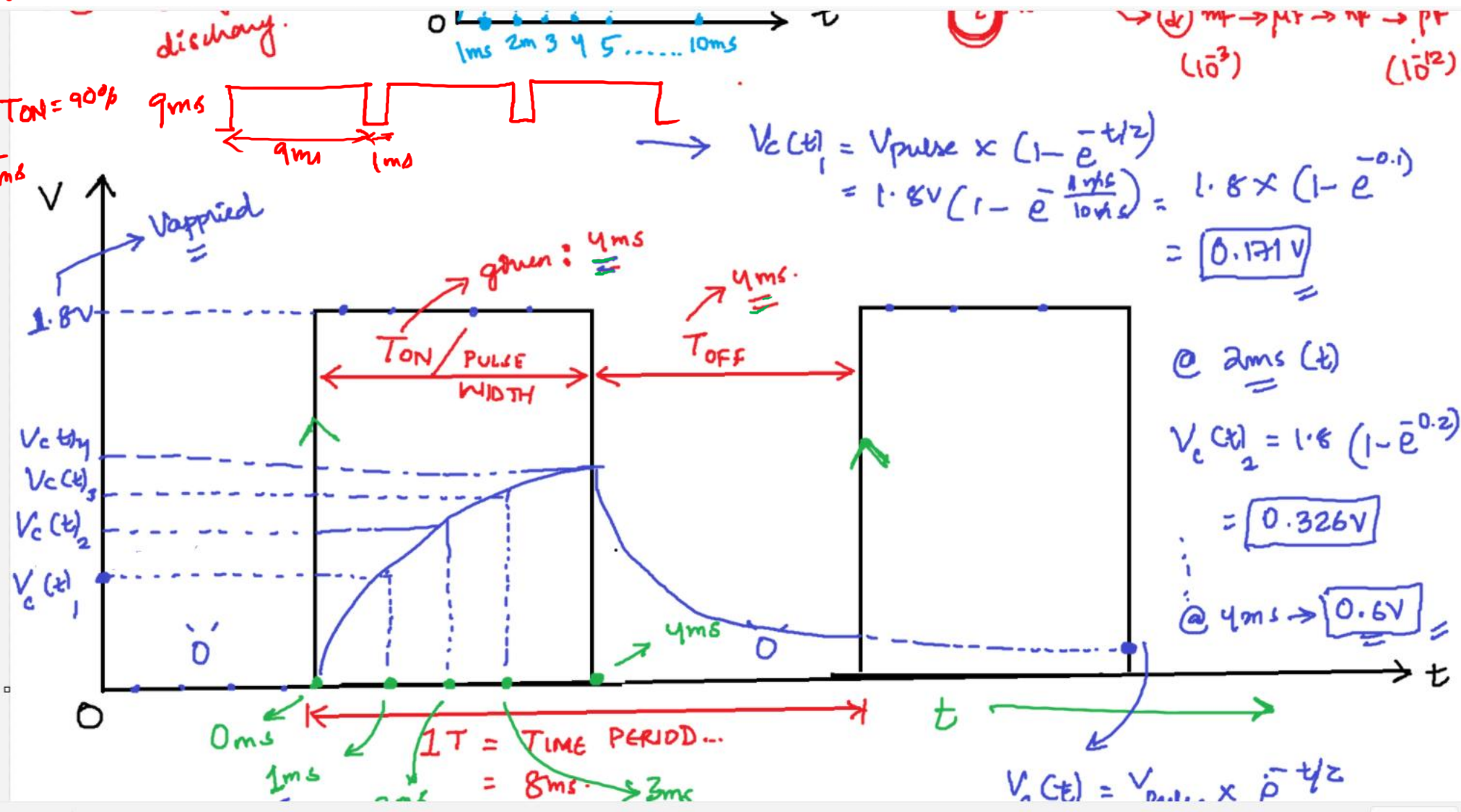
$T_{ON} = 50\%$ of total time period $= \frac{1}{2} \times 10\text{ms} = 5\text{ms}$



$T_{ON} = 80\%$



$T_{ON} = 90\%$



$$V_c(t)_1 = V_{\text{pulse}} \times (1 - e^{-t/\tau})$$

$$= 1.8\text{V} \left(1 - e^{-\frac{1\text{ms}}{10\text{ms}}}\right) = 1.8 \times (1 - e^{-0.1})$$

$$= \boxed{0.171\text{V}}$$

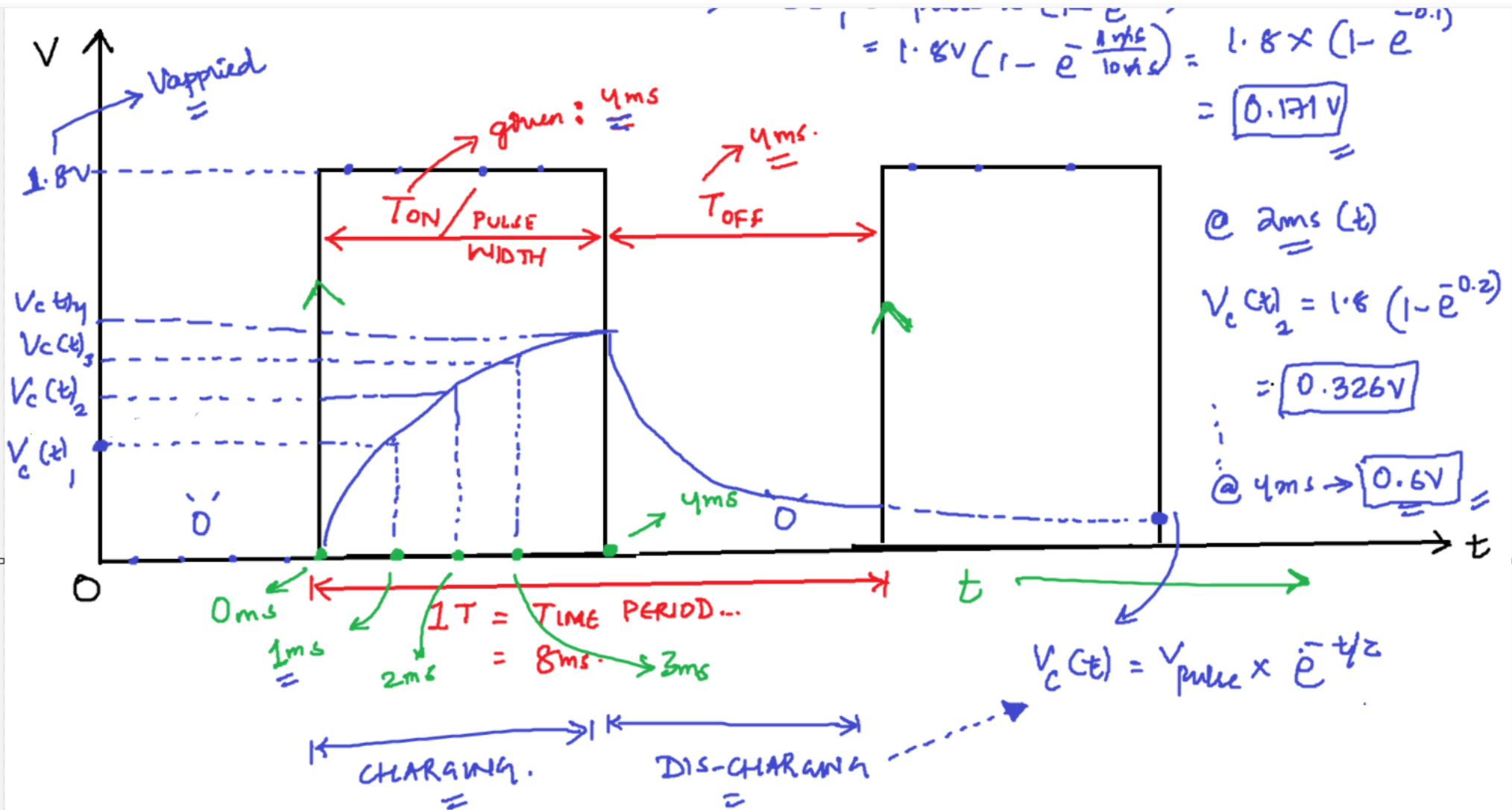
@ 2ms (t)

$$V_c(t)_2 = 1.8 (1 - e^{-0.2})$$

$$= \boxed{0.326\text{V}}$$

@ $4\text{ms} \rightarrow \boxed{0.6\text{V}}$

$$V_c(t) = V_{\text{pulse}} \times e^{-t/\tau}$$



CONCLUSION $\rightarrow \therefore T_{ON} = 4ms$

But @ 4ms ; Cap $\rightarrow 0.6V$ i.e. $\frac{1}{3}^{rd}$ of the Applied Voltage. (1.8V)

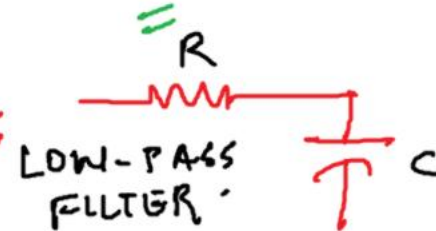
$\rightarrow$ Hence; it Needs Enough time to get CHARGED to its FULL POTENTIAL.

$\hookrightarrow$ This is NOTHING BUT DELAY !!

S_o; R-C ckt produces delays to the ckt.

$\boxed{*}$ ALSO ACTS AS A "FILTER"
 $\downarrow$
SS, DSP...

$R \rightarrow C$;



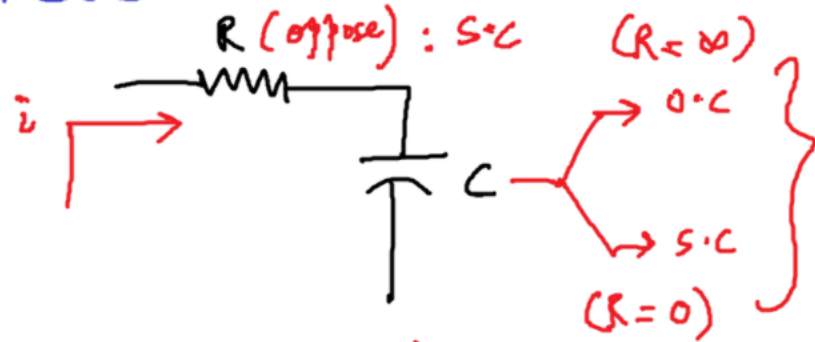
$C \rightarrow R$



CONCEPT OF LOW-PASS FILTER :-

R (oppose) : $S \propto C$ ($R < \infty$)

CONCEPT OF LOW-PASS FILTER :-



$t \rightarrow$
 $f = \frac{1}{t}$ ①



$\bar{R}' \rightarrow \infty \rightarrow O.C$
 $\bar{R}' \rightarrow 0 \rightarrow S.C$

PASS
 FILTER.

$R_c = X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$

- D.C $\rightarrow f = 0 \text{ Hz} \rightarrow 0 \text{ Hz}$
- A.C $\rightarrow 50-60 \text{ Hz} = \infty \text{ Hz}$

$f = \infty$

$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{\infty} = 0$

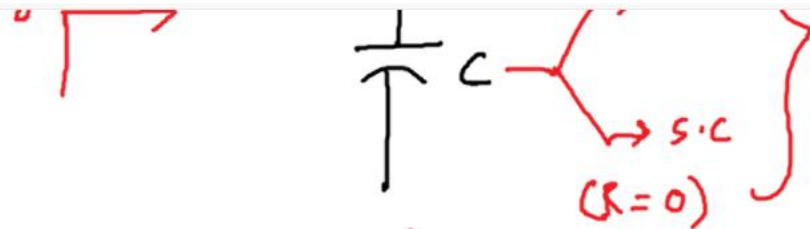
$R = 0 \rightarrow S.C$

$f = 0$

$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{0} = \infty$

$R = \infty \rightarrow O.C$

$\rightarrow$ HIGH PASS } A-C Analysis.



$$\left. \begin{array}{l} R \rightarrow \infty \rightarrow \text{O.C.} \\ R' \rightarrow 0 \rightarrow \text{S.C.} \end{array} \right\} R_c = X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

- D.C $\rightarrow f = 0 \text{ Hz} \rightarrow 0 \text{ Hz}$
- A.C $\rightarrow 50-60 \text{ Hz} = \infty \text{ Hz}$

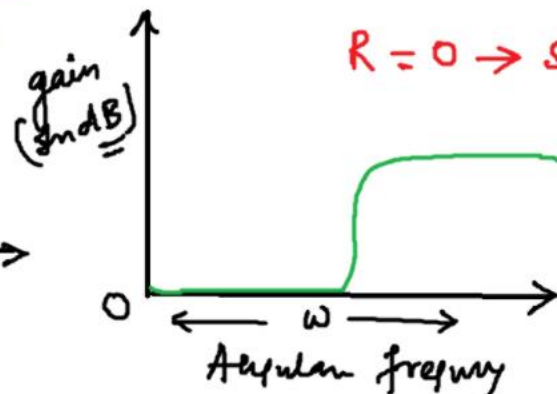
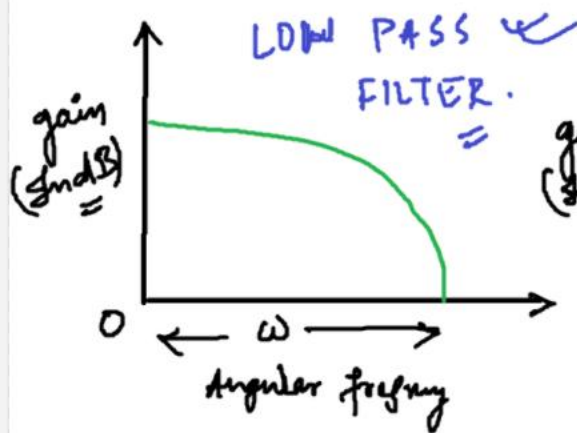
$t \rightarrow$
 $f = \frac{1}{t}$ ①

$$f = \infty$$

$$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{\infty} = 0$$

$$f = 0$$

$$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{0} = \infty$$



$\rightarrow$ HIGH PASS FILTER

A-C Analysis
 AC amplitude ✓
 phase = 0° ✓

TIME CONSTANT VARIATION

[14 Cases from $R = 1 \, \underline{\underline{\Omega}}$ to $1 \underline{\underline{G}} \, \underline{\underline{\Omega}}$]

$C = 10 \underline{\underline{n}} \, \underline{\underline{F}}$ to $1 \underline{\underline{f}} \, \underline{\underline{F}}$]

$\downarrow \uparrow \quad Z = R \cdot C \quad \downarrow \quad 10^9 \rightarrow 10^{-15}$

CASE 01 :

$R = 1k\ \Omega$; $C = 10f\ F$

Launch File Edit View Create Check Options Window Parasitics/LDE ADE Explorer Calibre XKit Utils XT018 Utils PVS Help

Basic

Replace (None) Sim Time 0

Navigator

Schematic rc_ckt_dc

OBJECTS

All

Instances 5

Nets 4

Pins 2

Nets and Pins

GROUPS

Cells

Types

+

Property Editor

Schematic Attributes

Library sailesh

Cell rc_ckt_dc

View schematic

Mode editable

Last Saved Sat Feb 8 ...

Locked By vlab14@td...

Units inch

mouse L: schSingleSelectPt()

M: hiZoomAbsoluteScale(getCurrentWindow() 0.9)

R: schHiMousePopUp()

1(2)

[Terminal]

[Virtuoso® 6.1.8-64b - Log: /home/...]

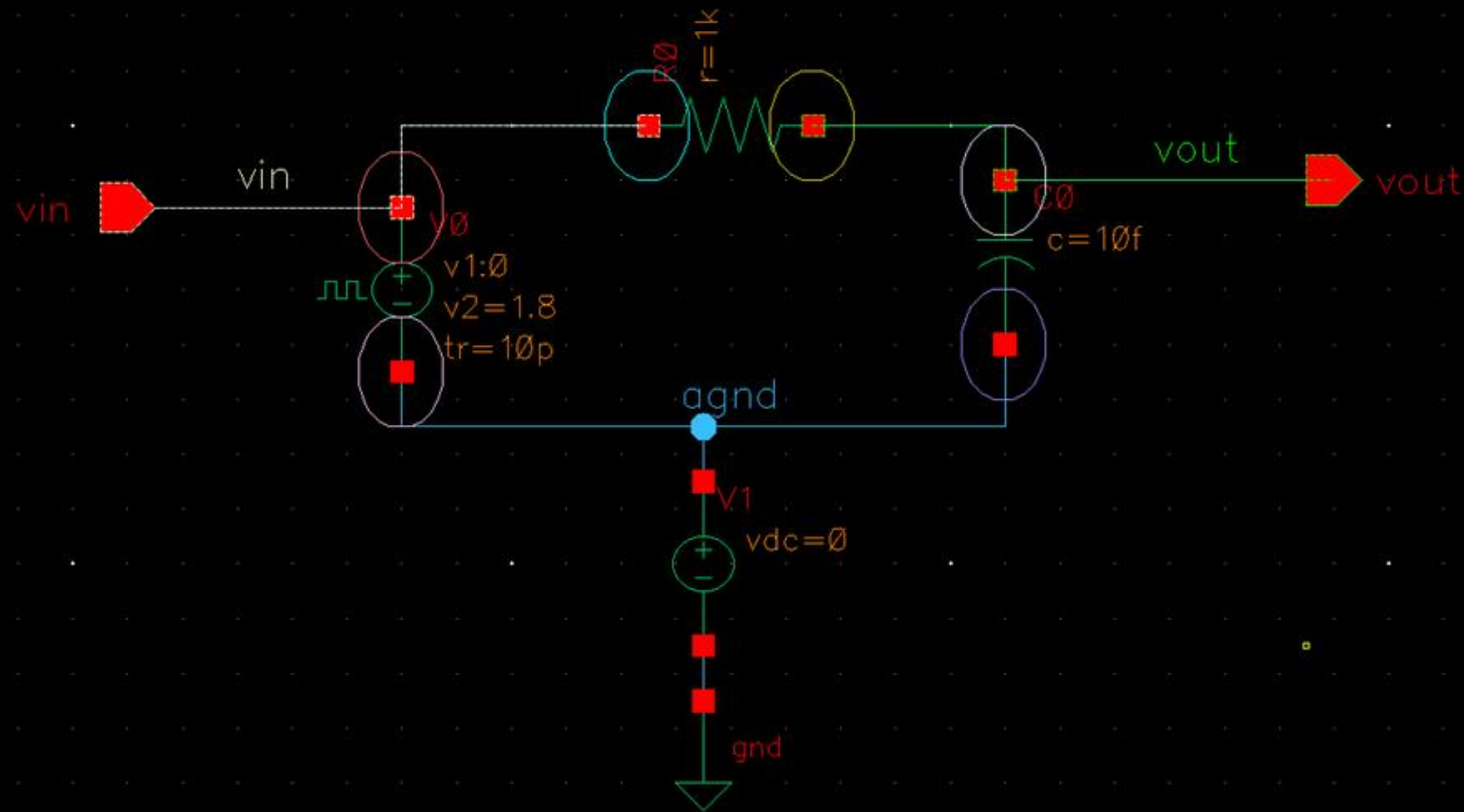
[Library Manager: Directory ...ainings...]

[Virtuoso® ADE Explorer Editing: sai...]

[/home/local/simulation/vlab14/saile...]

[Virtuoso (R) Visualization & Analysis...]

$$\tau = R \cdot C$$





sailesh_rc_ckt_dc_1

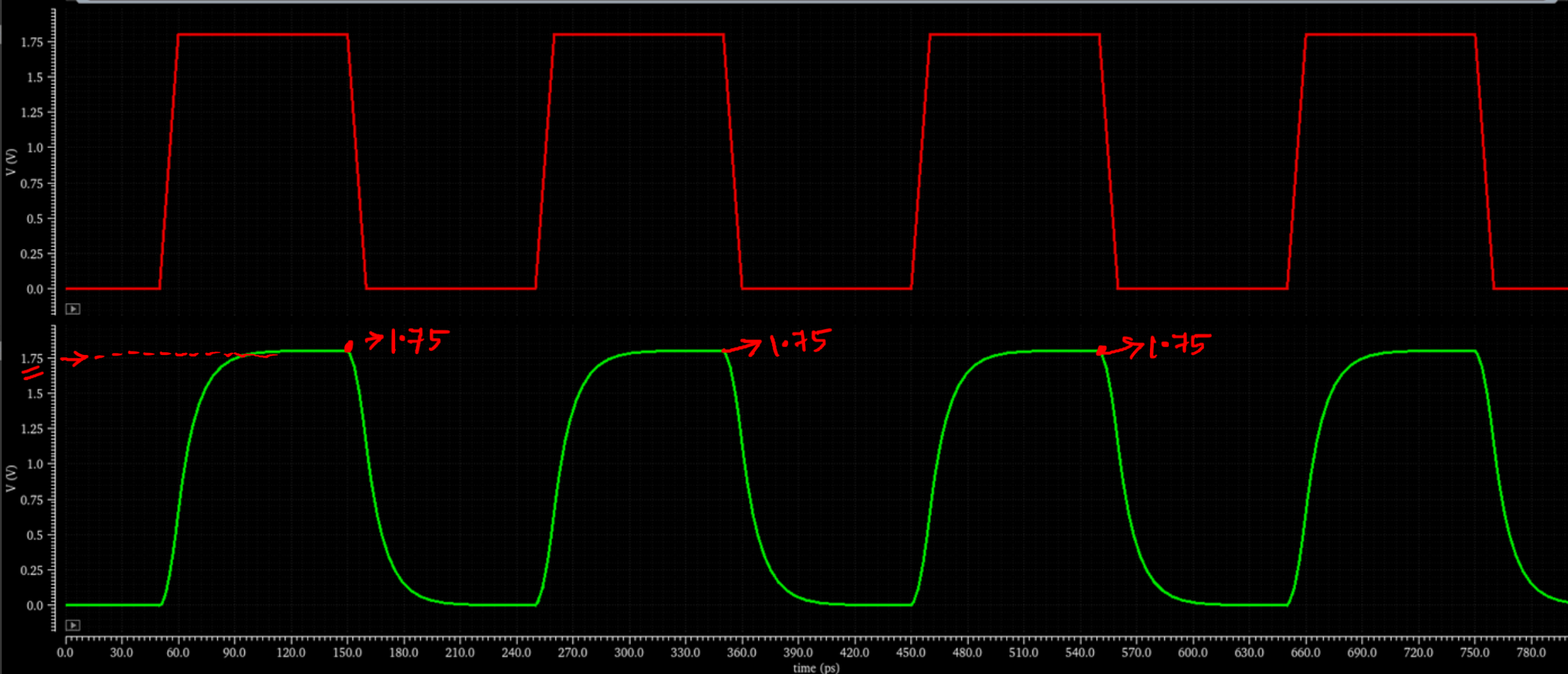
Transient Response

Sat Feb 8 10:57:13 2025

Name Vis

/vin

/vout



mouse L:

M:

R:

6(20) no error



sailesh_rc_ckt_dc_1

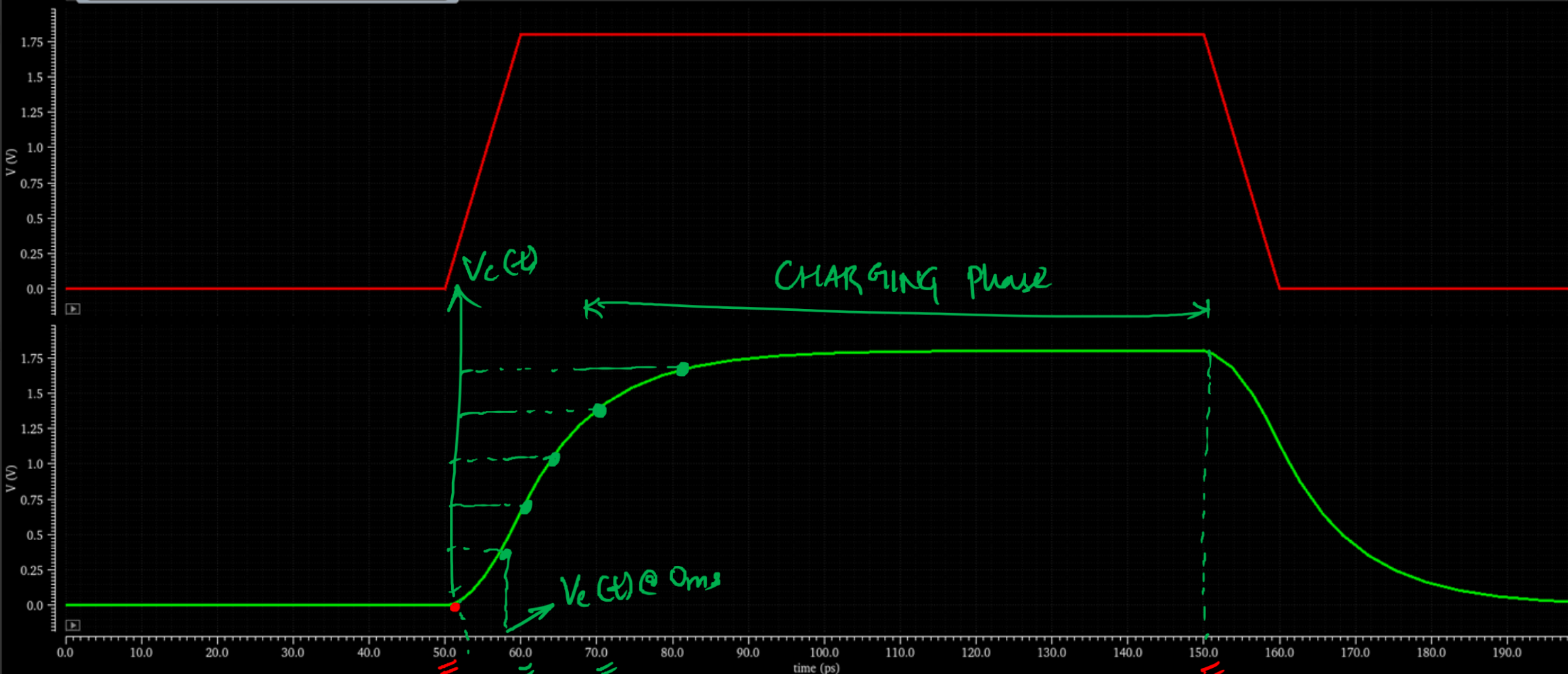
Transient Response

Sat Feb 8 10:57:13 2025

Name Vis

/vin

/vout

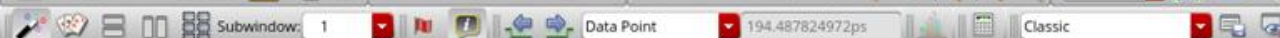


mouse L:

M:

R:

6(20) Trace: Avin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



sailesh_rc_ckt_dc_1

Transient Response

Sat Feb 8 10:57:13 2025 1



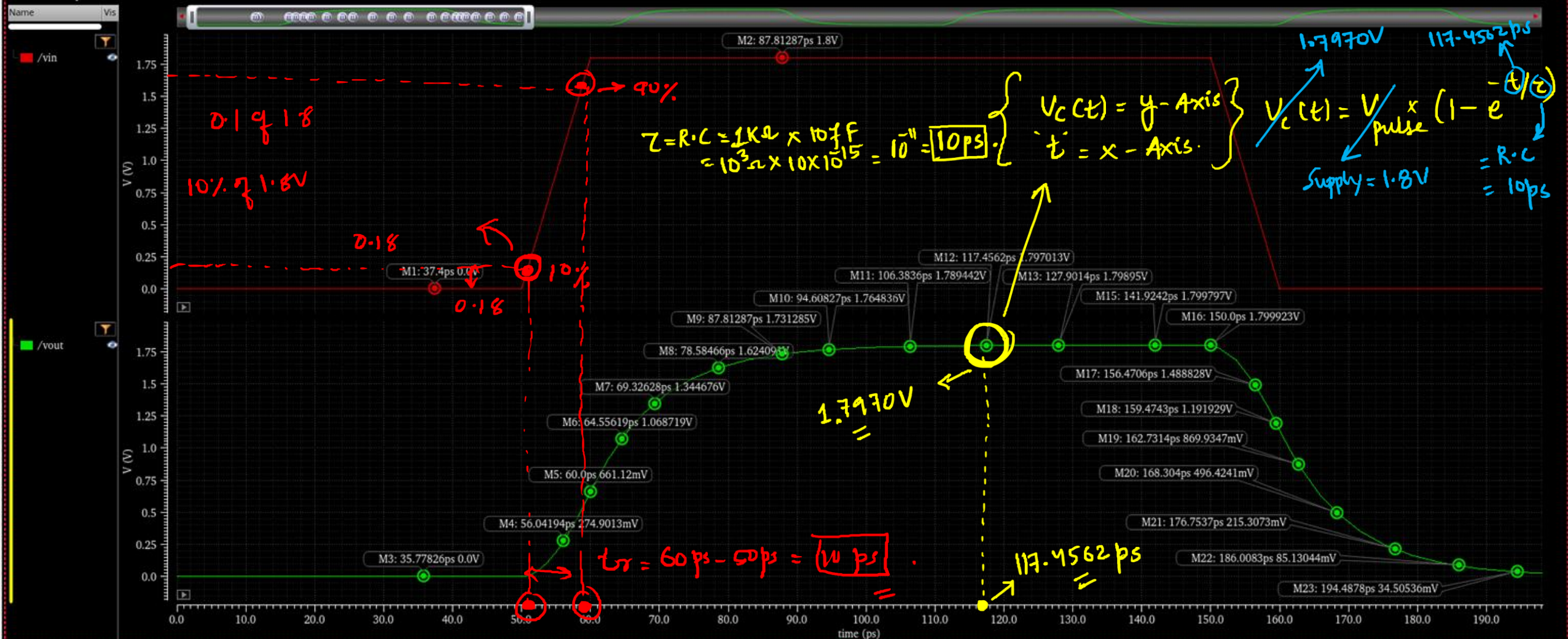
mouse L:

M:

6(20) Trace: Avout; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom

Transient Response

Sat Feb 8 10:57:13 2025 1



mouse L:

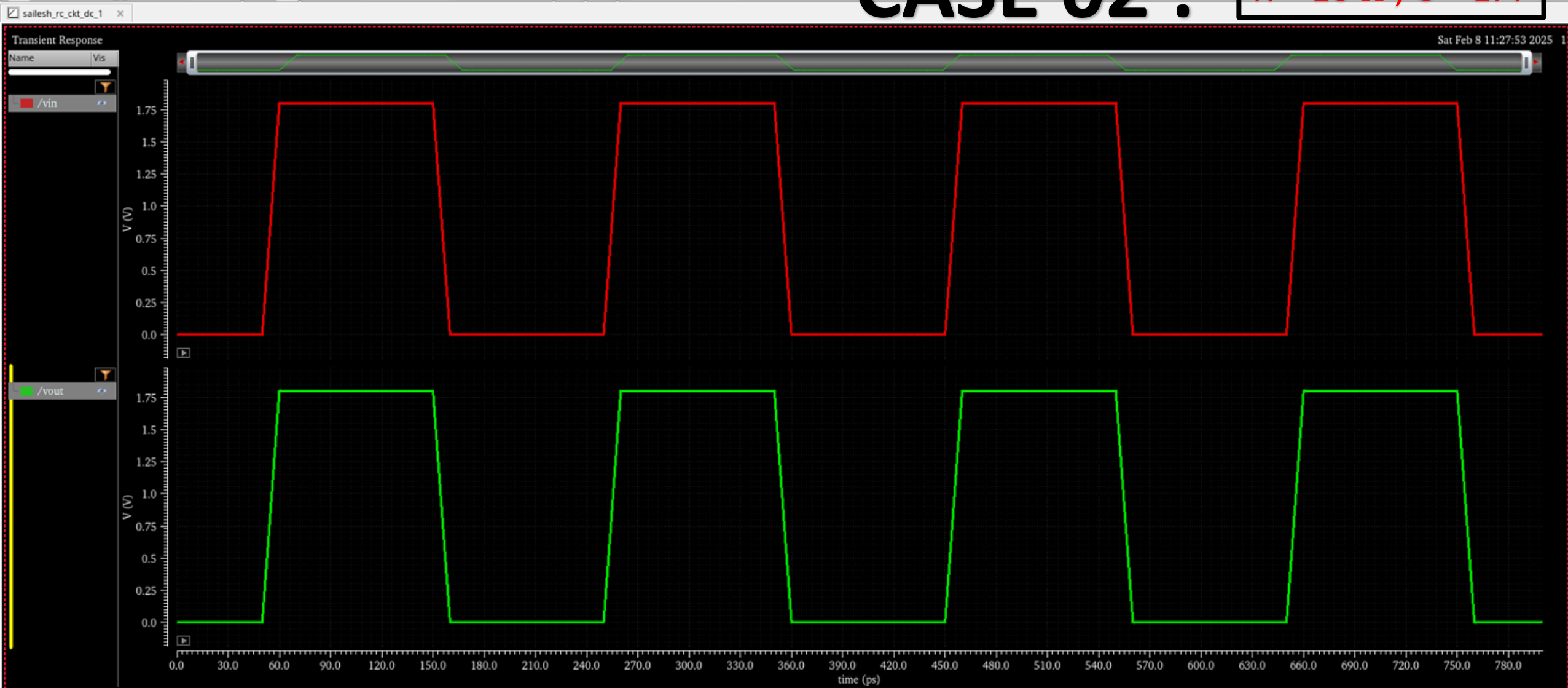
M:

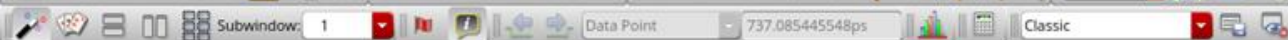
6(20) Trace: Avout; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



CASE 02 :

$R = 10\ \Omega$; $C = 1\text{fF}$





sailesh_rc_ckt_dc_1

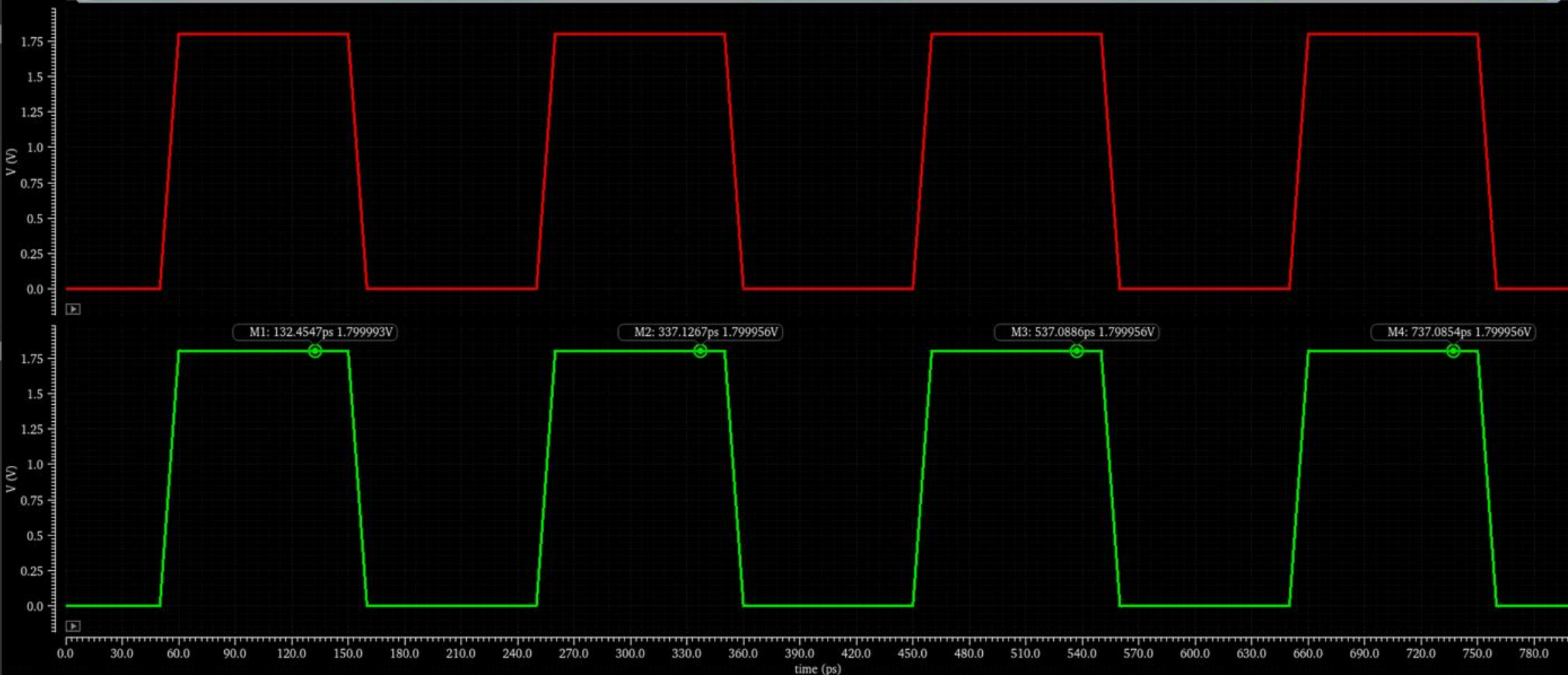
Transient Response

Sat Feb 8 11:27:53 2025 1

Name Vis

/vin

/vout

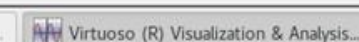


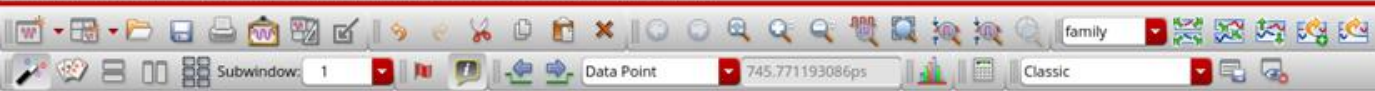
mouse L:

M:

R:

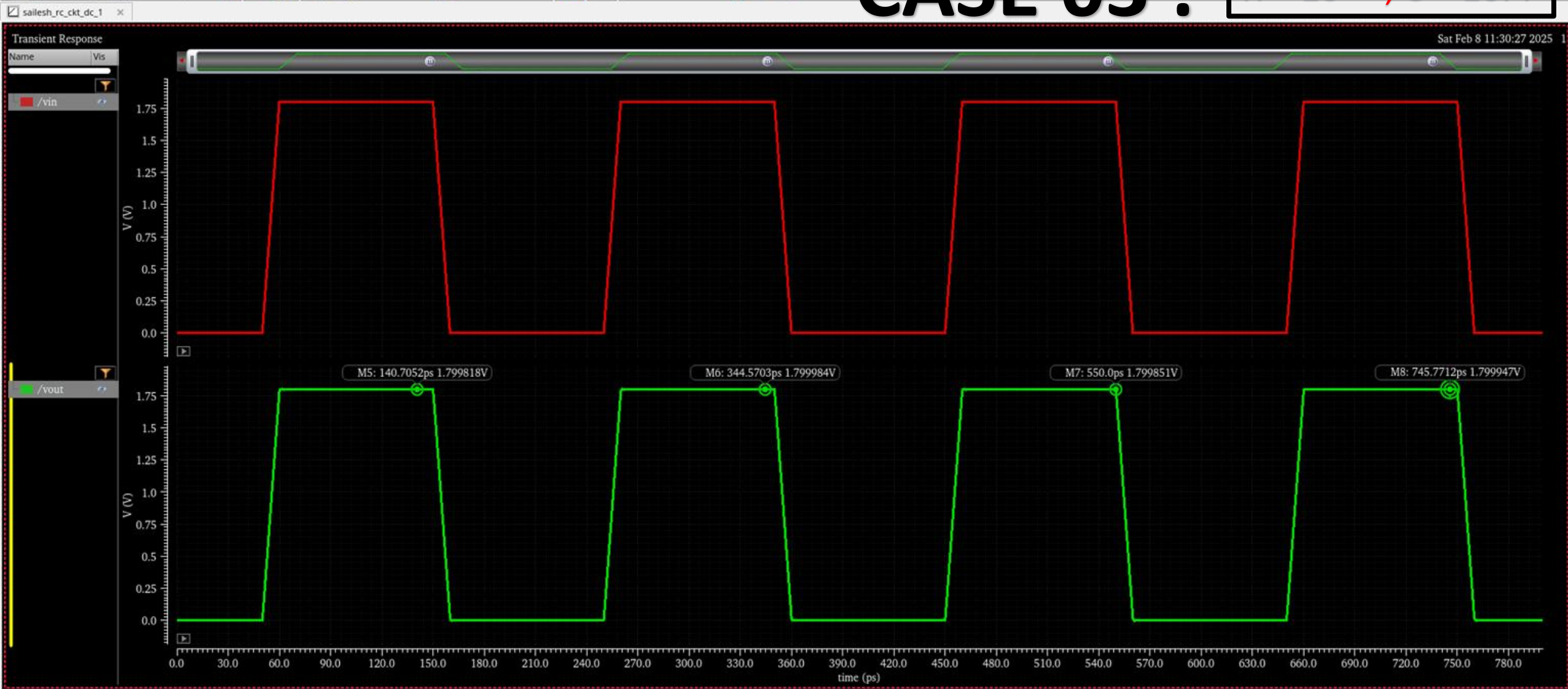
4(11) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



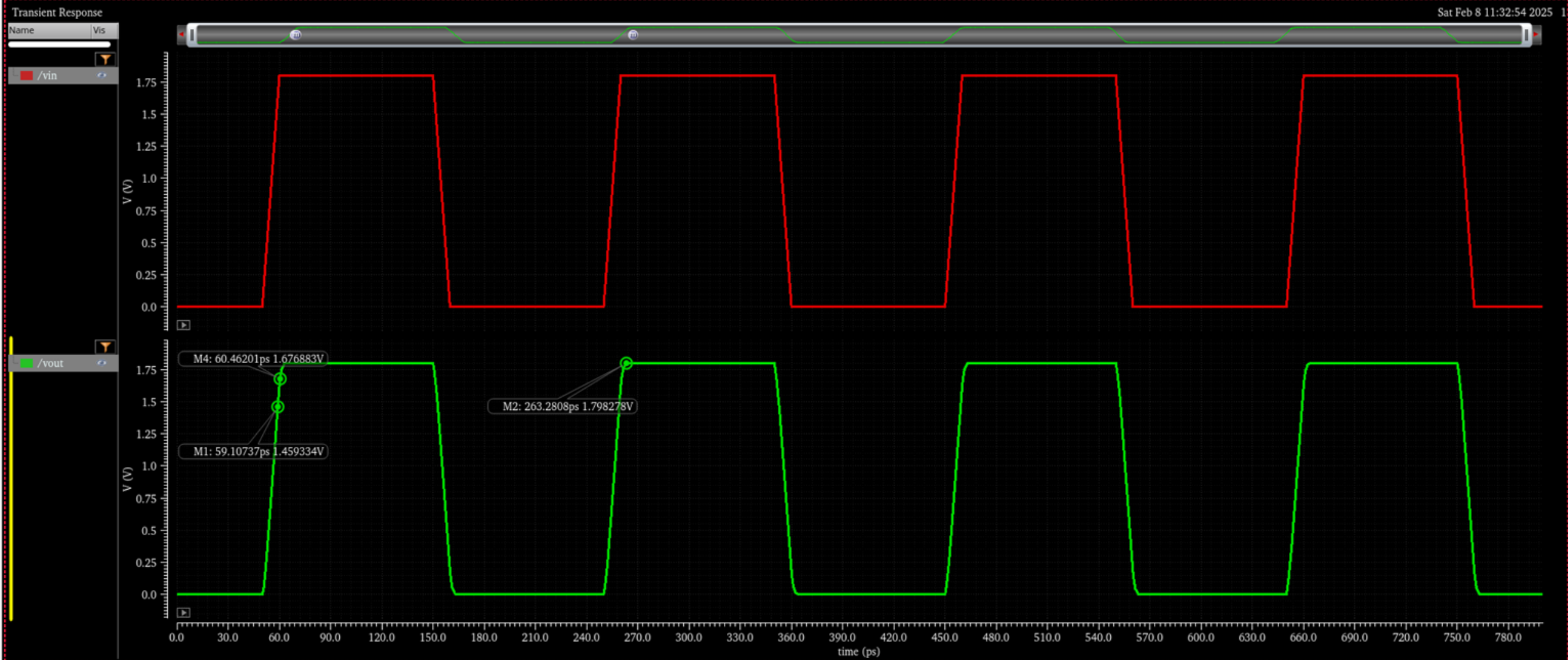


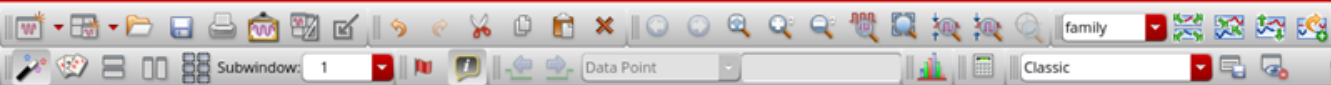
CASE 03 :

$R = 10\ \Omega$; $C = 10\text{fF}$



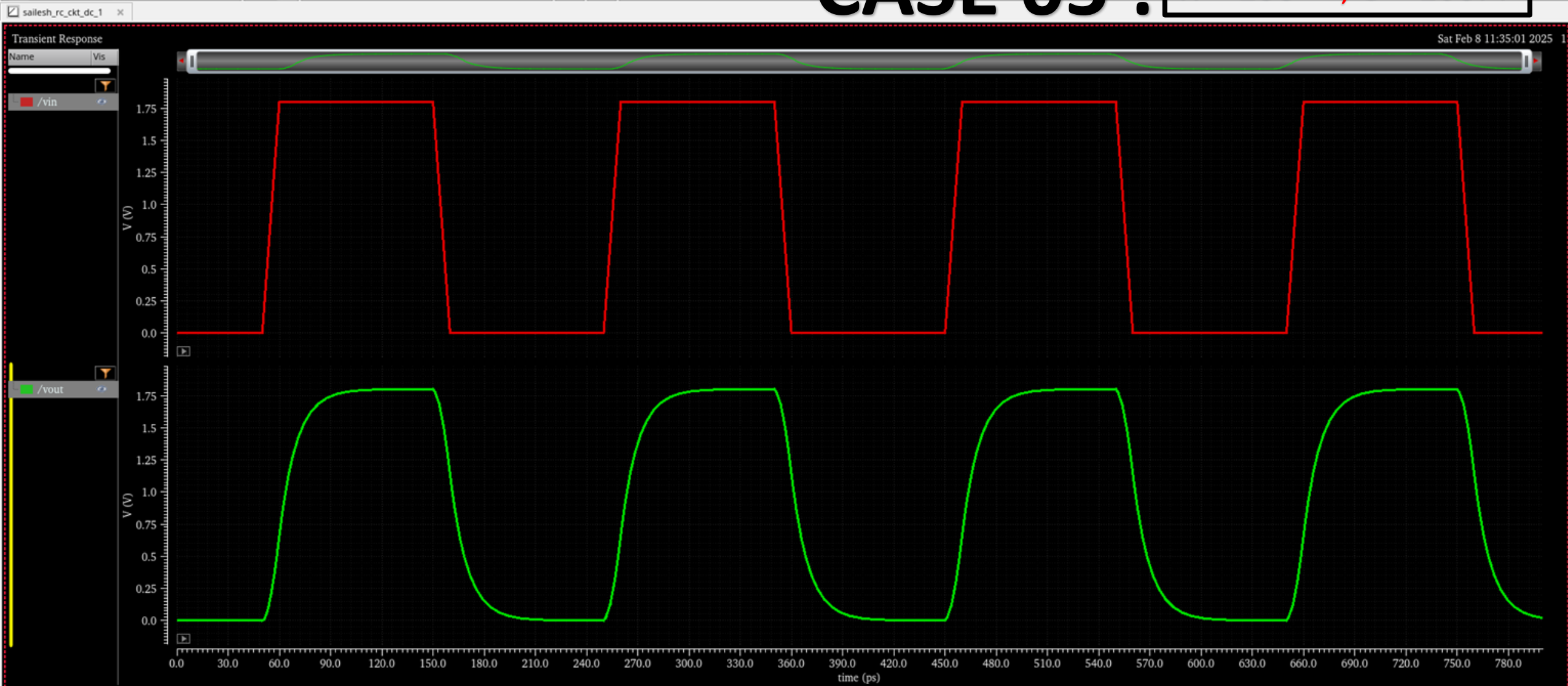
CASE 04 : $R = 100\ \Omega$; $C = 10\text{fF}$

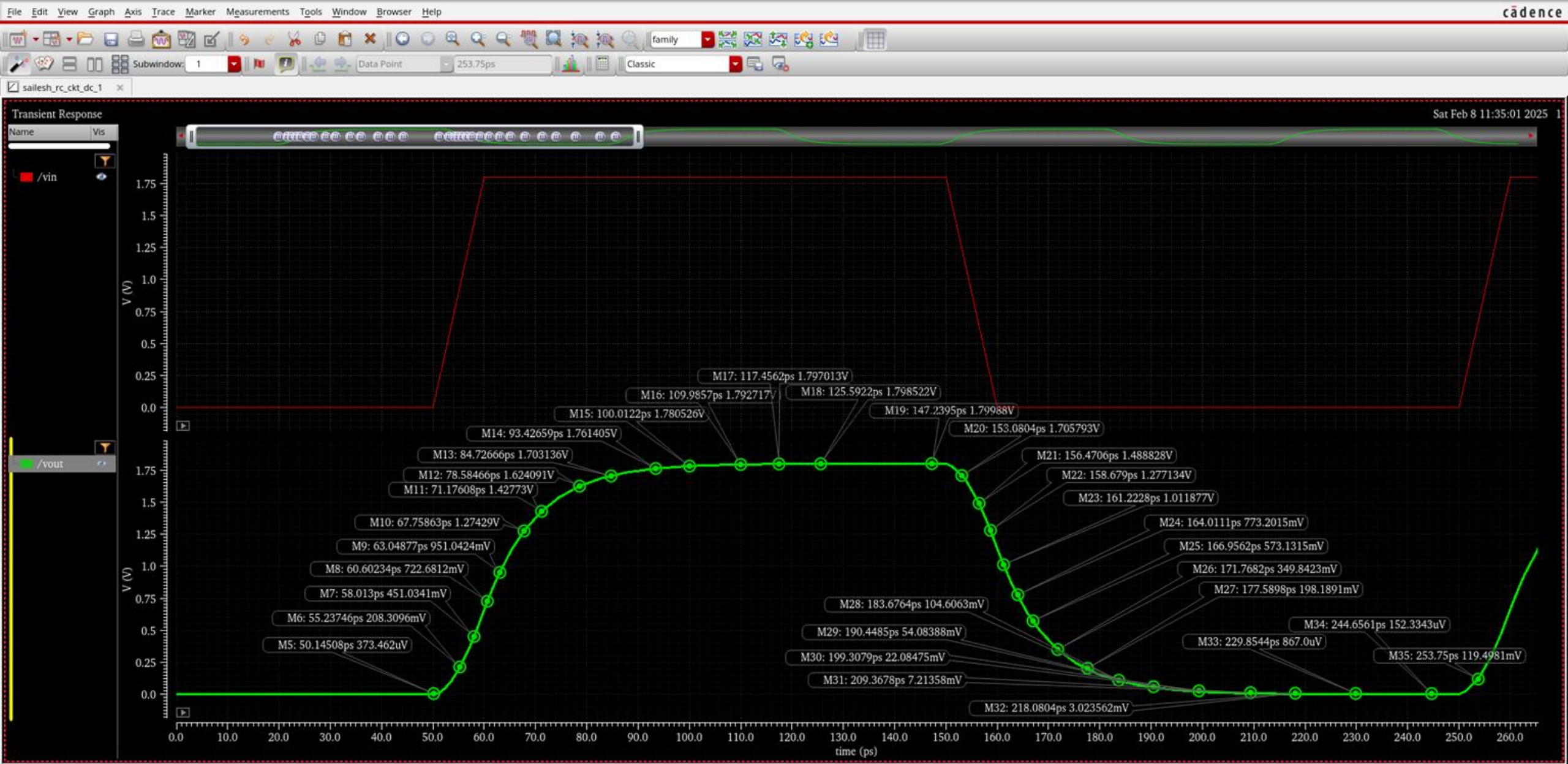




CASE 05 :

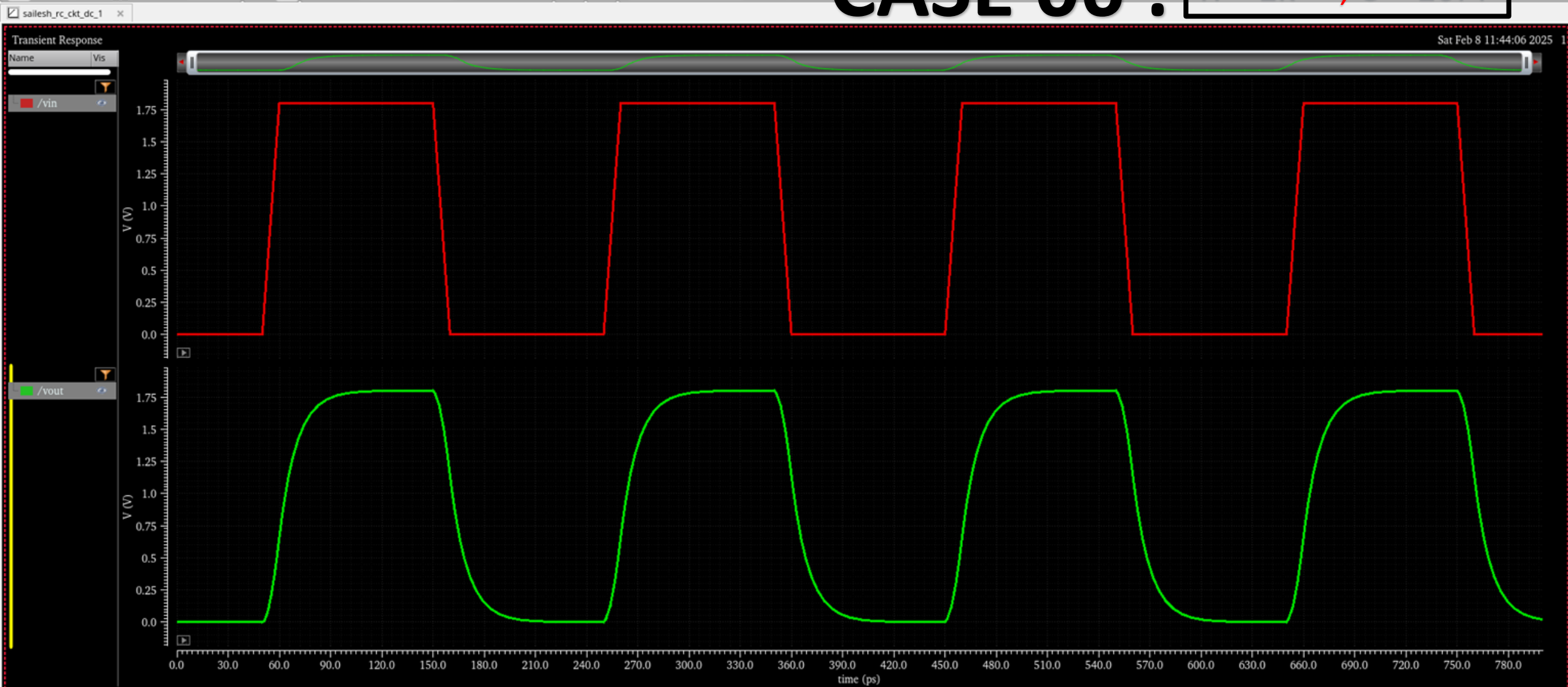
$R = 100 \Omega$; $C = 100f F$







CASE 06 : $R = 1k\ \Omega$; $C = 10f\ F$



Virtuoso (R) Visualization & Analysis XL: sailesh rc_ckt_dc maestro



File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

cadence



Subwindow: 1 Data Point: 258.226576163ps Classic

sailesh_rc_ckt_dc_1

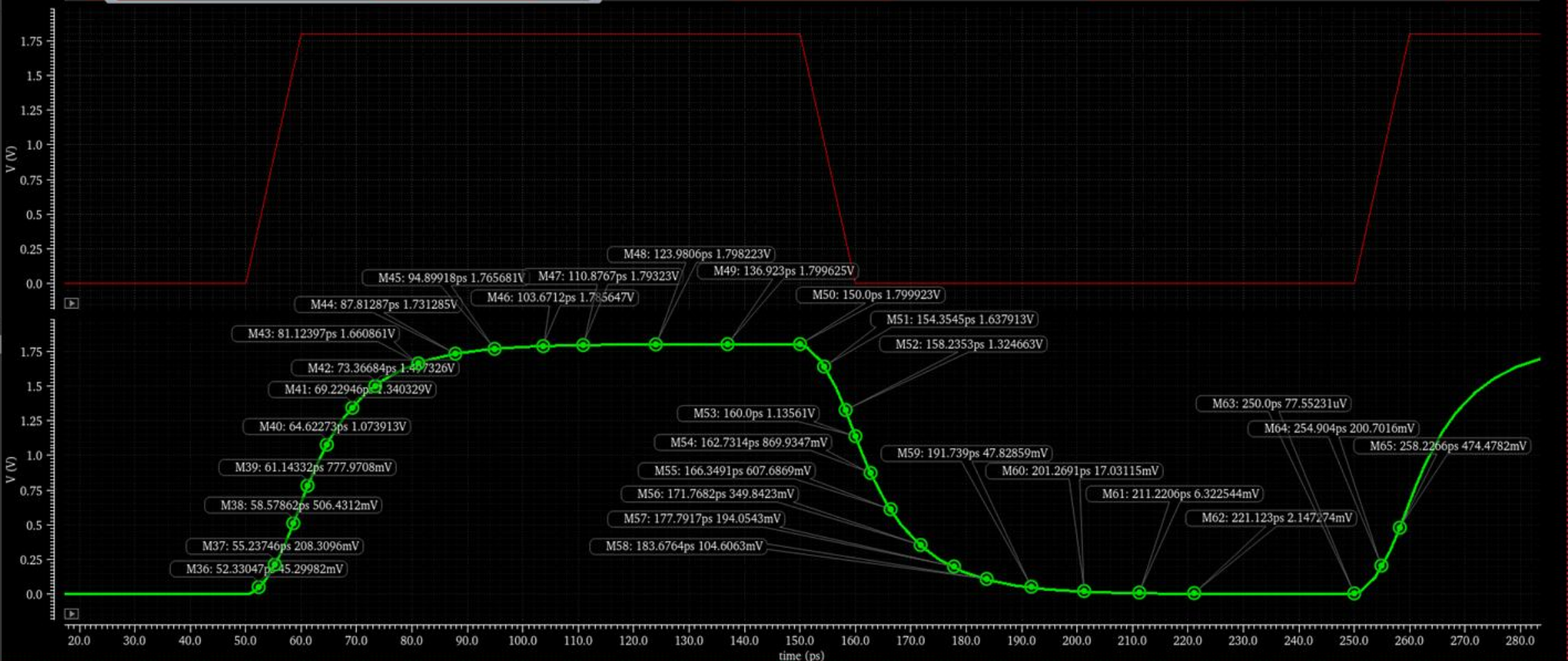
Transient Response

Sat Feb 8 11:44:06 2025 1

Name Vis

/vin

/vout



mouse L:

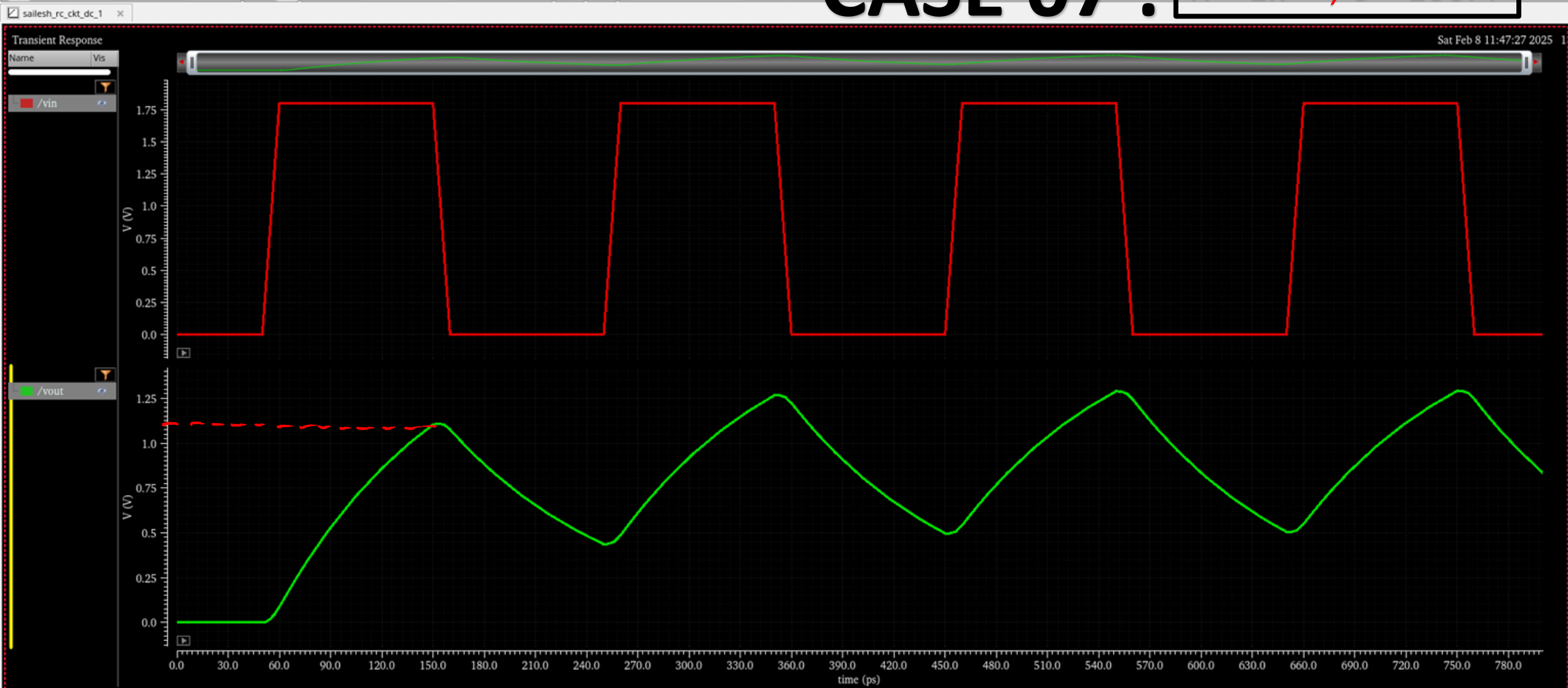
M:

R:

8(24) Trace: /vout; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



CASE 07 : $R = 1k\ \Omega$; $C = 100f\ F$





sailesh_rc_ckt_dc_1

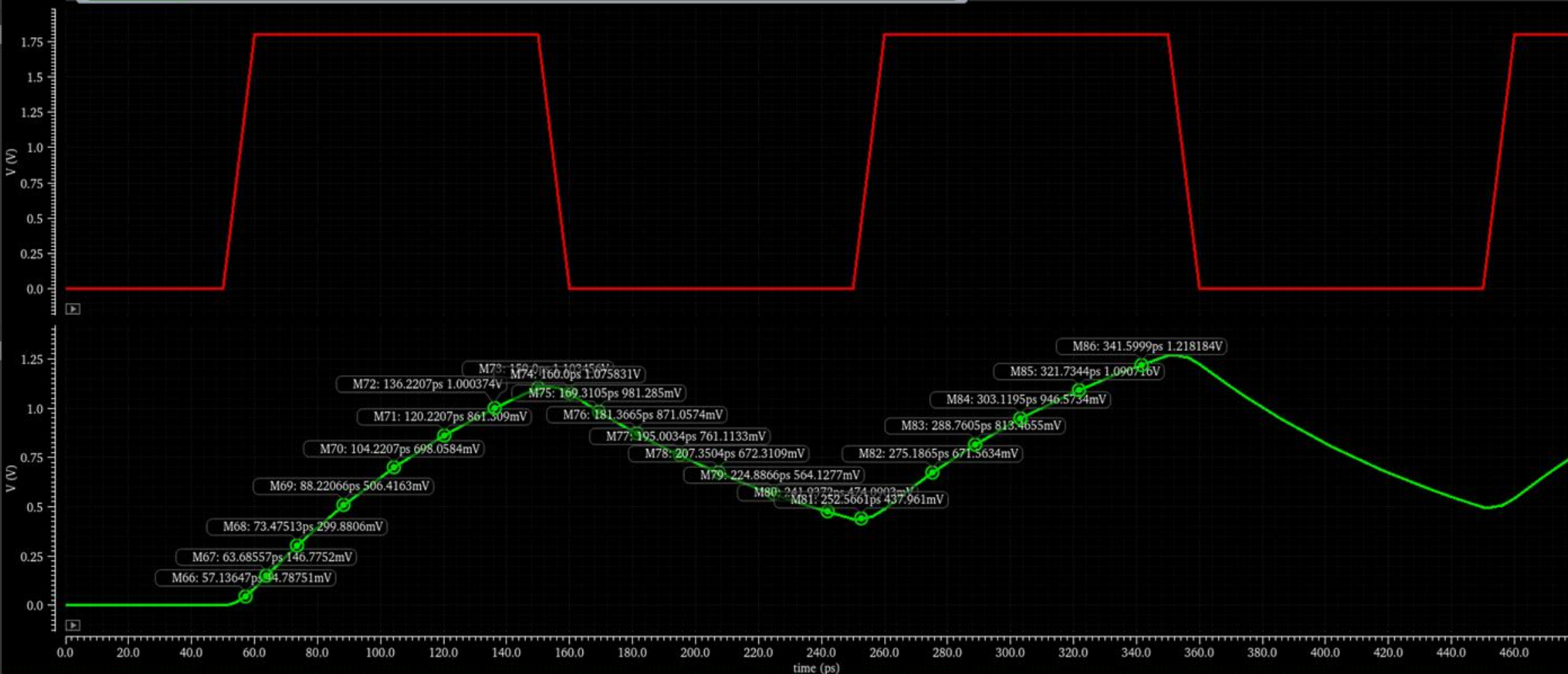
Transient Response

Sat Feb 8 11:47:27 2025 1

Name Vis

/vin

/vout

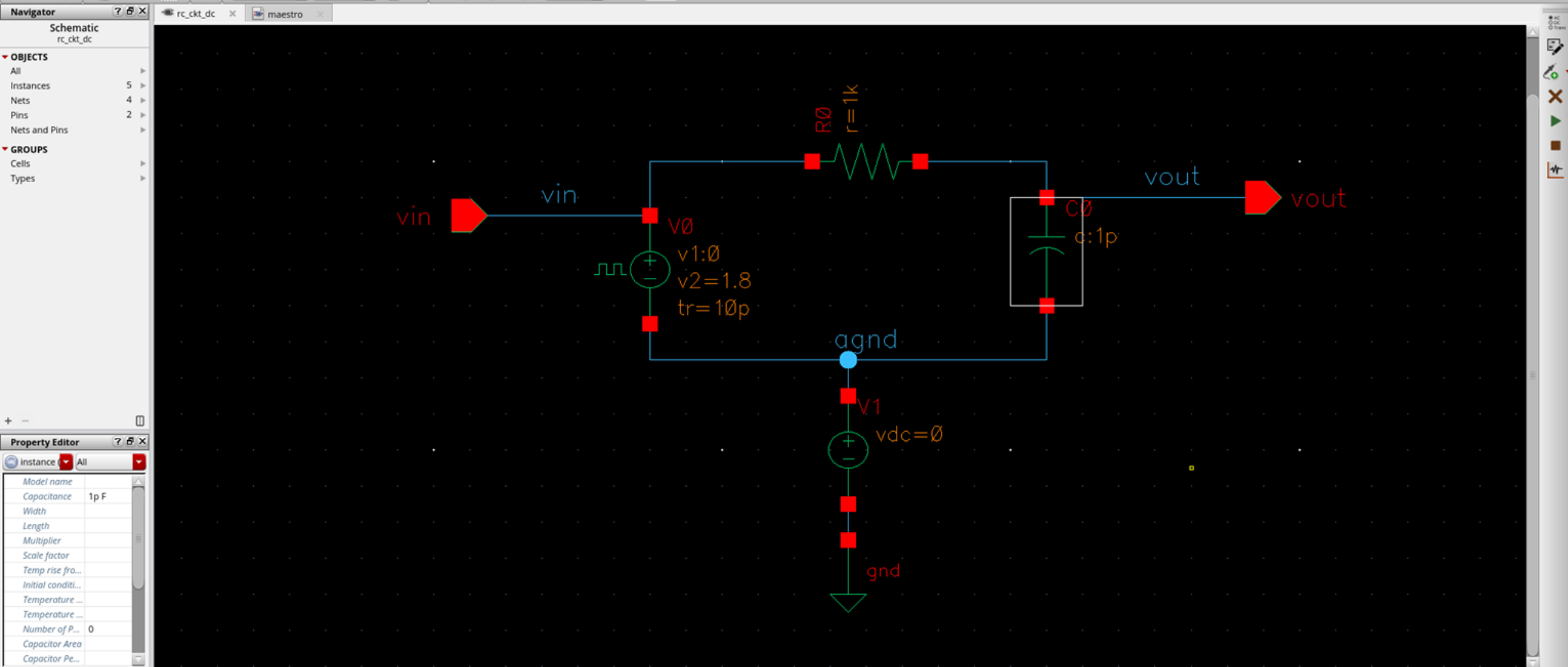


mouse L:

M:

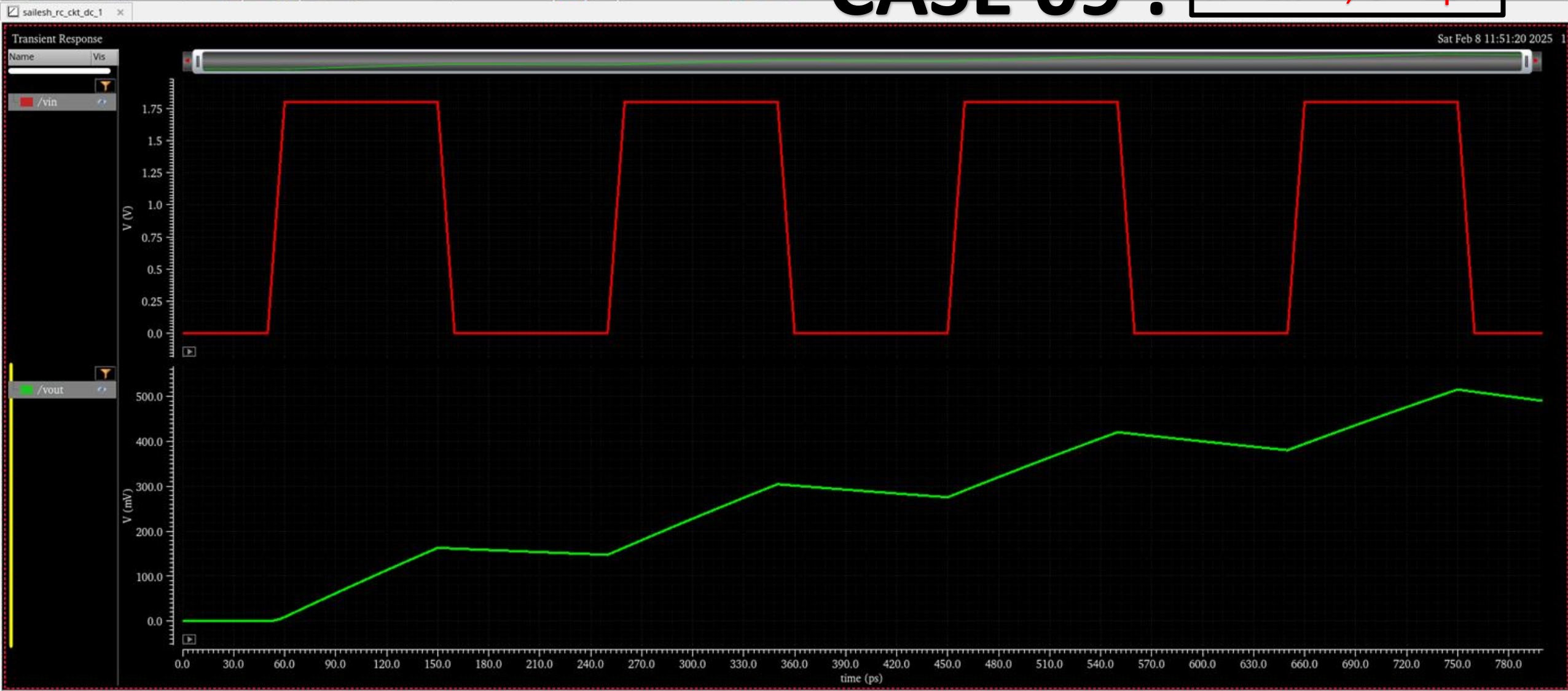
R:

9(27) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom





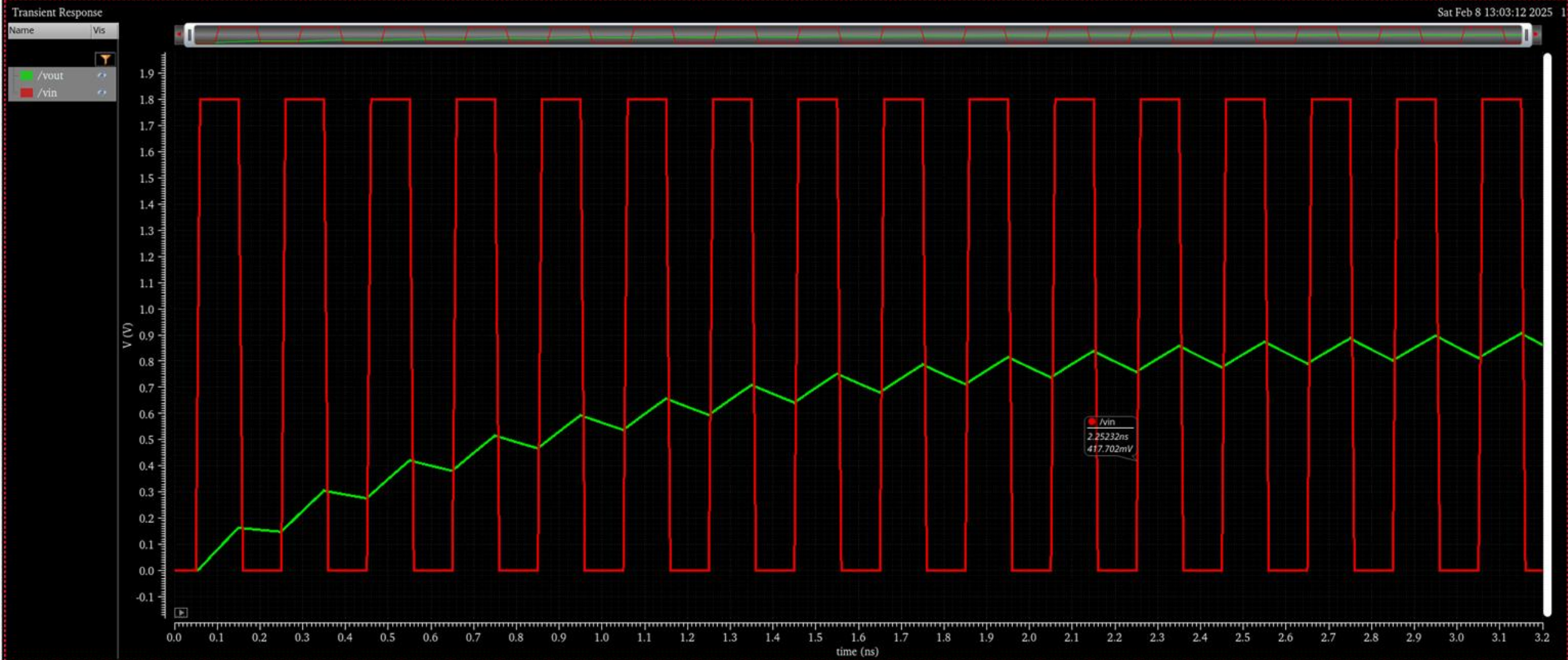
CASE 09 : $R = 1k\ \Omega$; $C = 1p\ F$



ACTUAL GRAPH OF CASE 08 :

Showing Exponential growth ($e^{t/R.C}$)

$R = 1k\ \Omega$; $C = 1p\ F$



23(70) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom

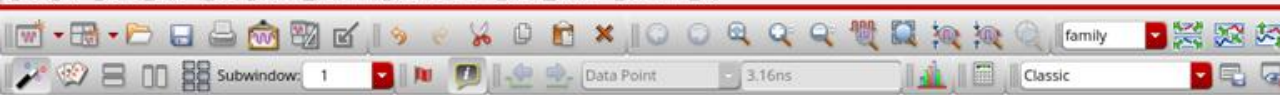
[Terminal] [Virtuoso® 6.1.8-64b - Log: /home/...]

[Library Manager: Directory ...ainings...]

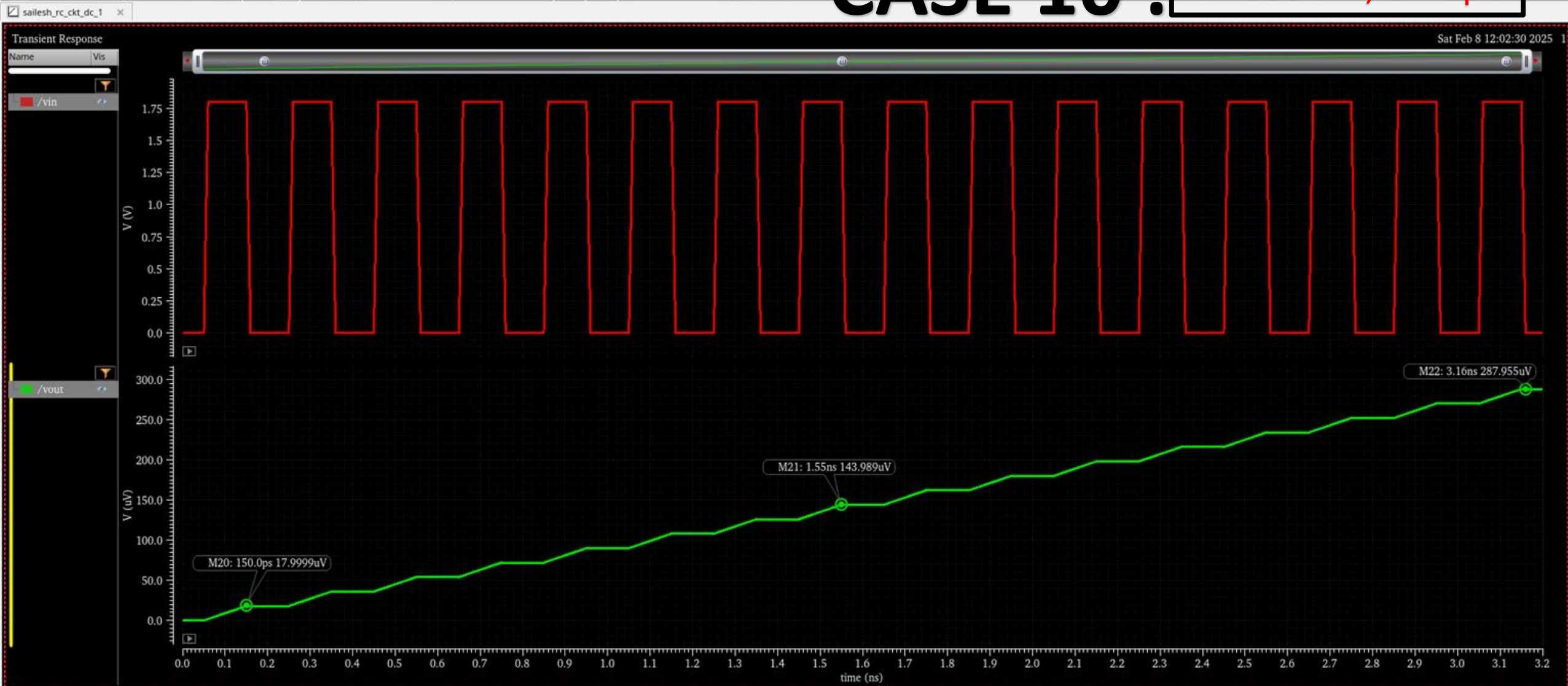
[Virtuoso® ADE Explorer Editing: sail...]

[/home/local/simulation/vlab14/saile...]

[Virtuoso (R) Visualization & Analysis...]

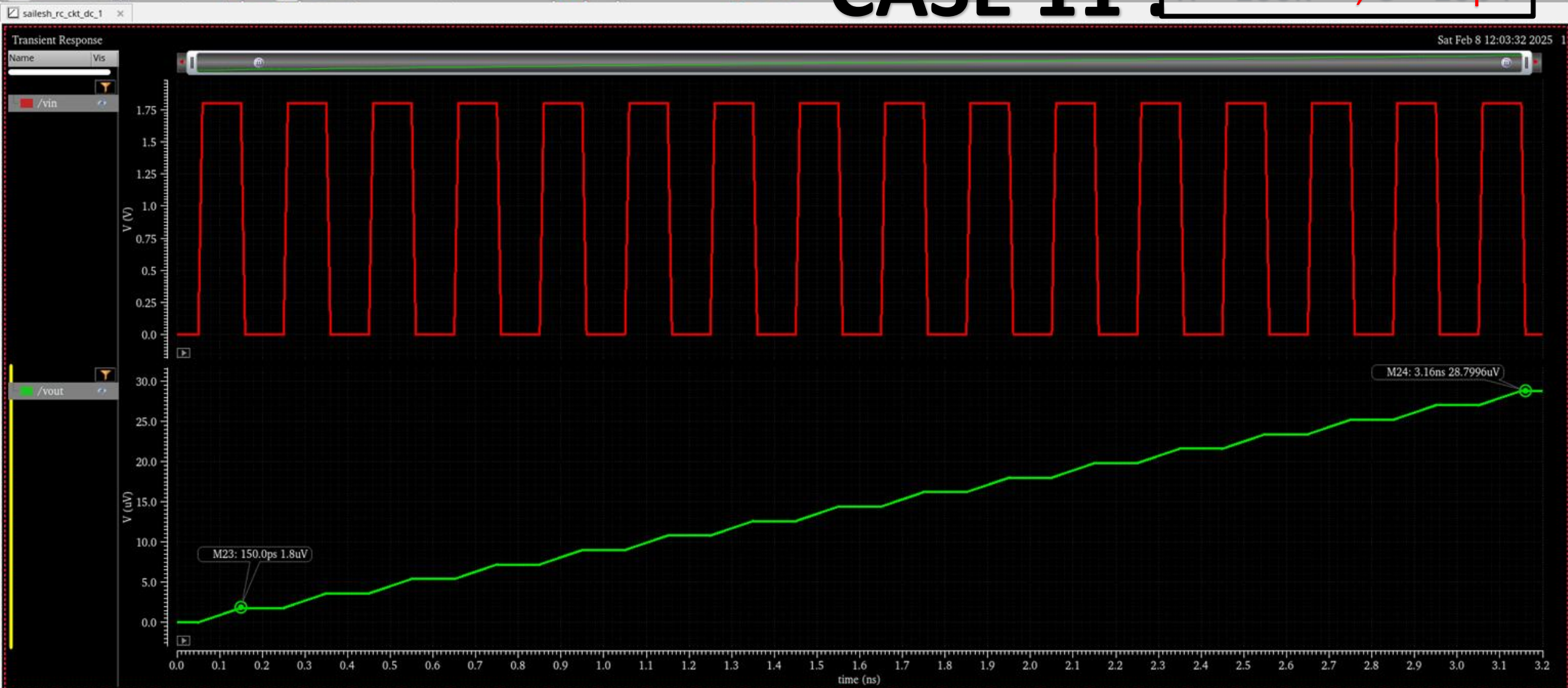


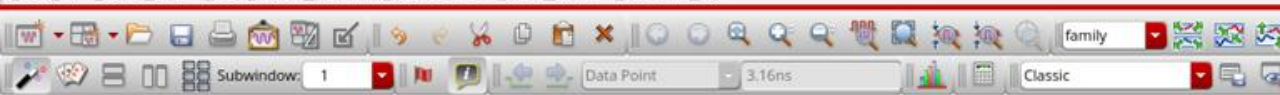
CASE 10 : $R = 100k\ \Omega$; $C = 1p\ F$



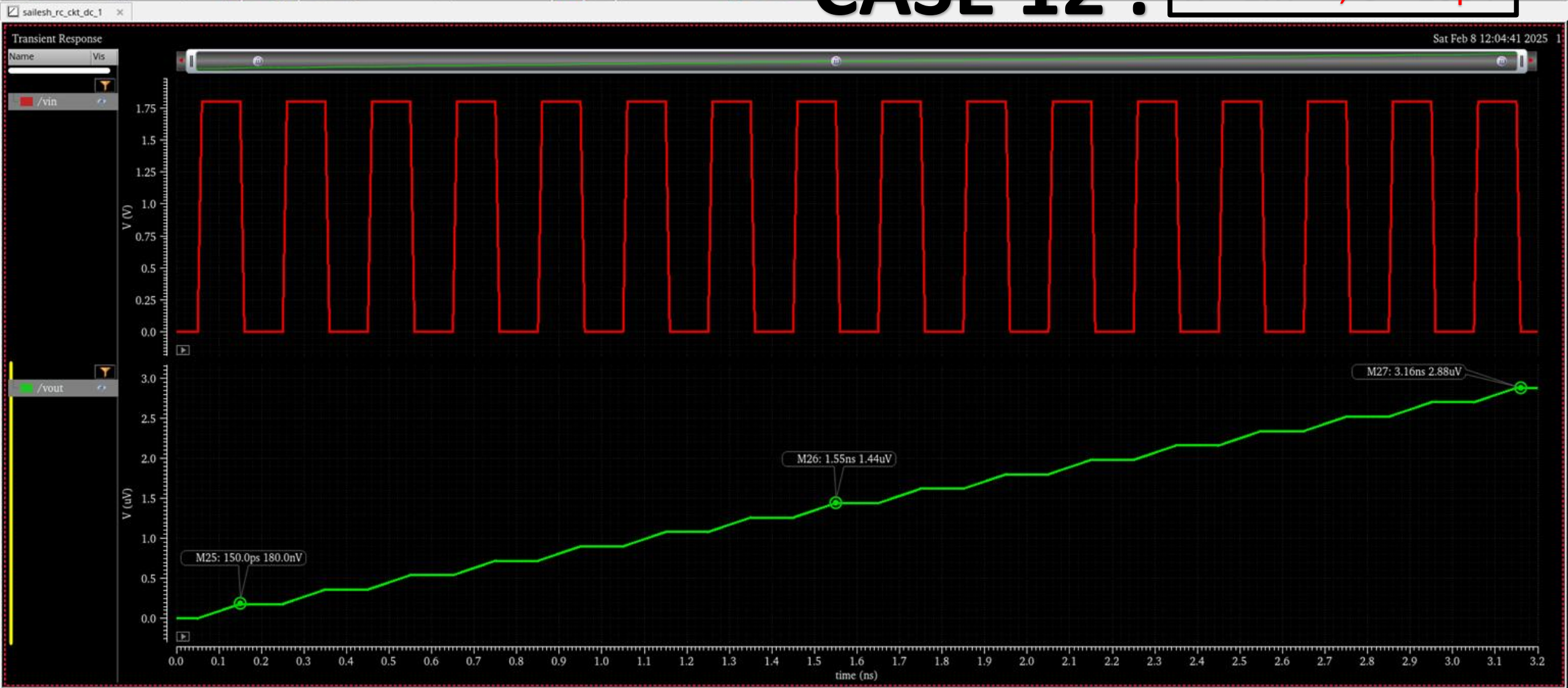


CASE 11 : $R = 100k \Omega$; $C = 10p F$

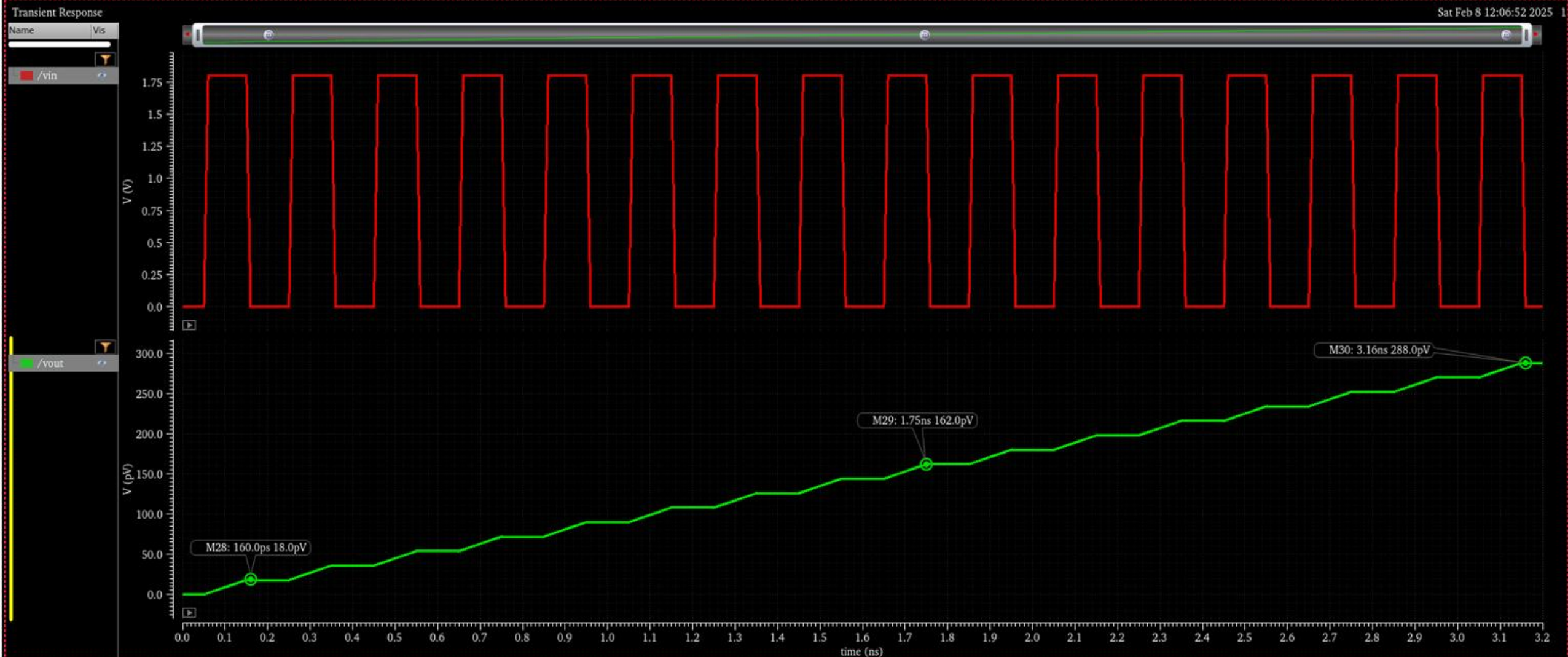




CASE 12 : $R = 1\text{M } \Omega$; $C = 10\text{p F}$

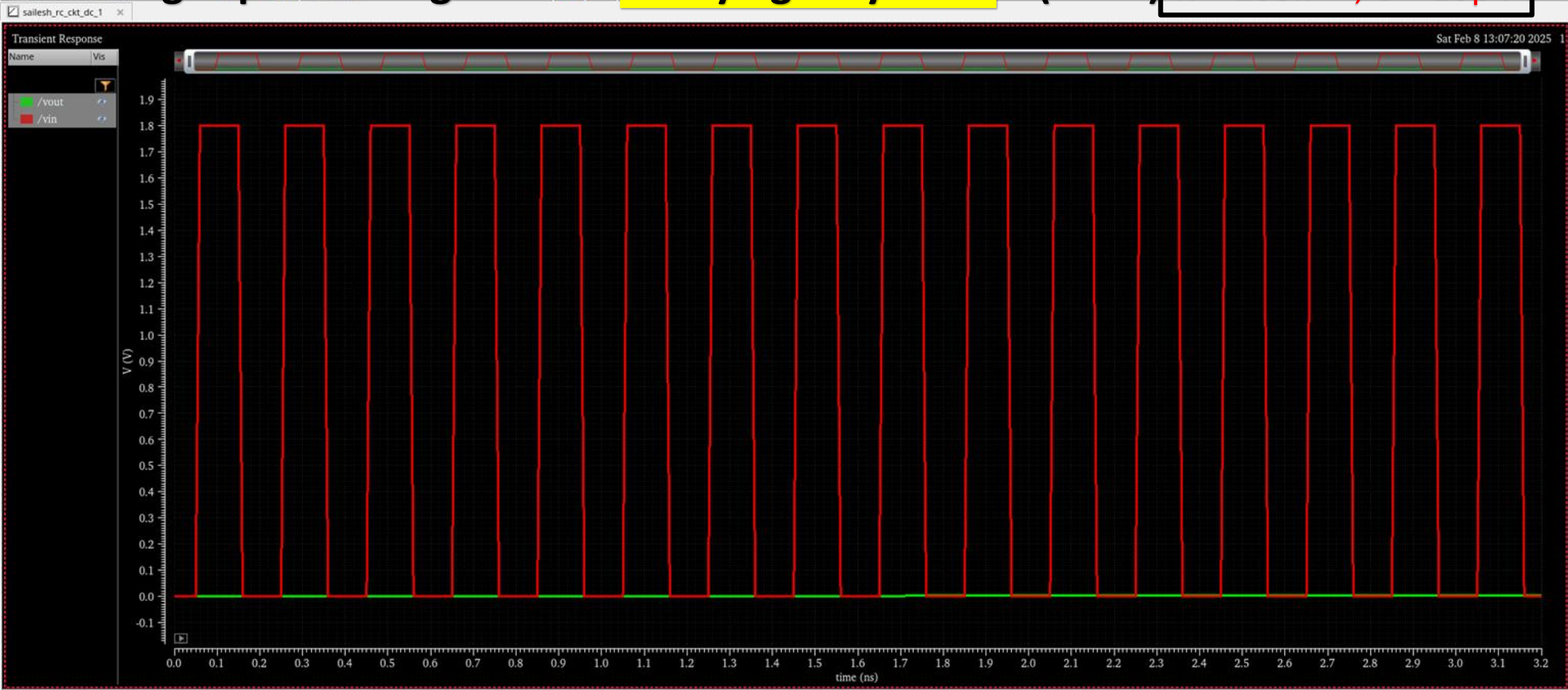


CASE 13 : $R = 1G \Omega$; $C = 10n F$



ACTUAL GRAPH OF CASE 11:

Showing Exponential growth & Decaying very FAST !! ($e^{t/R.C}$) R = 100k Ω ; C = 10p F



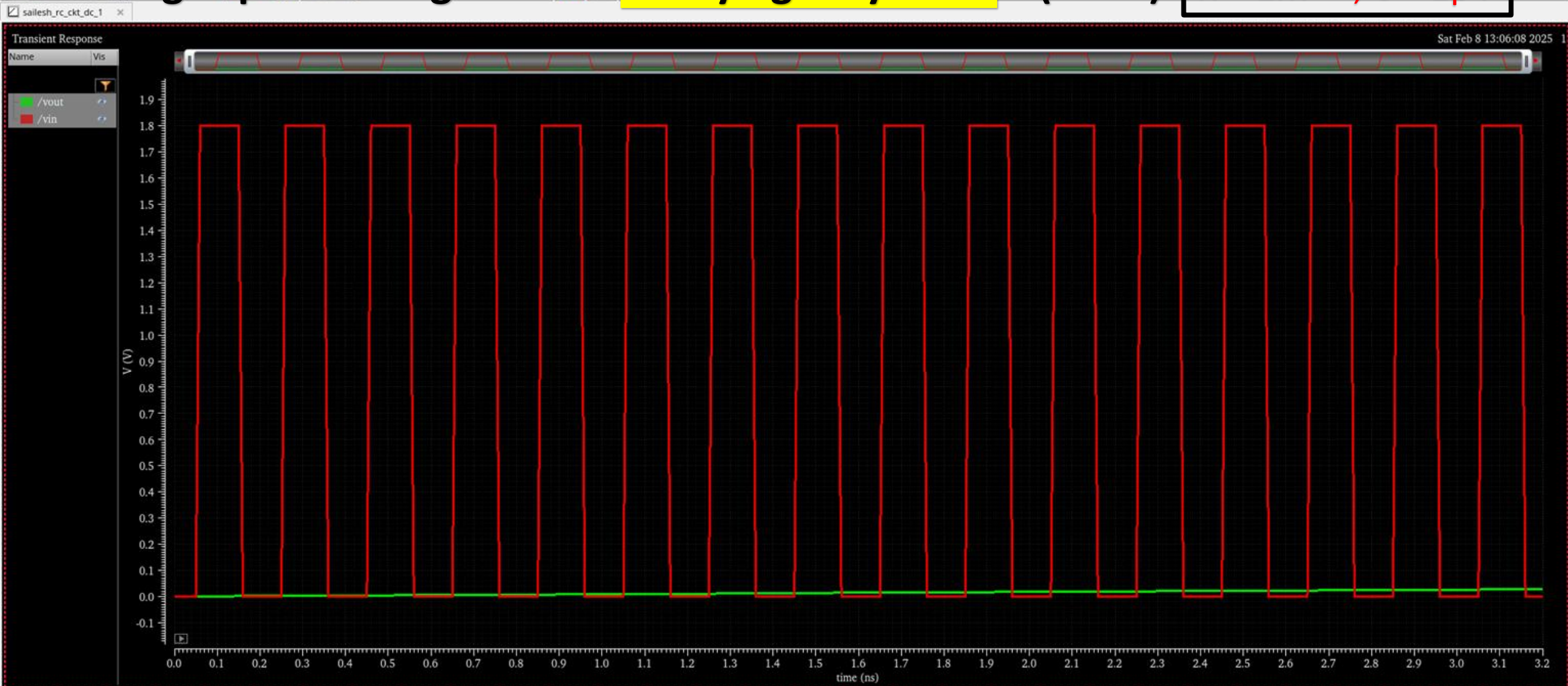
ACTUAL GRAPH OF CASE 12:

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

cadence

Showing Exponential growth & Decaying very FAST !! ($e^{t/R.C}$)

$R = 1M \Omega ; C = 1p F$



mouse L:

M:

R:

25(76) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom

SUDDEN CHARGING & INSTANTANEOUSLY DISCHARGING!!! CASE 14 :

 $R = 1\text{M}\Omega ; C = 1\text{nF}$

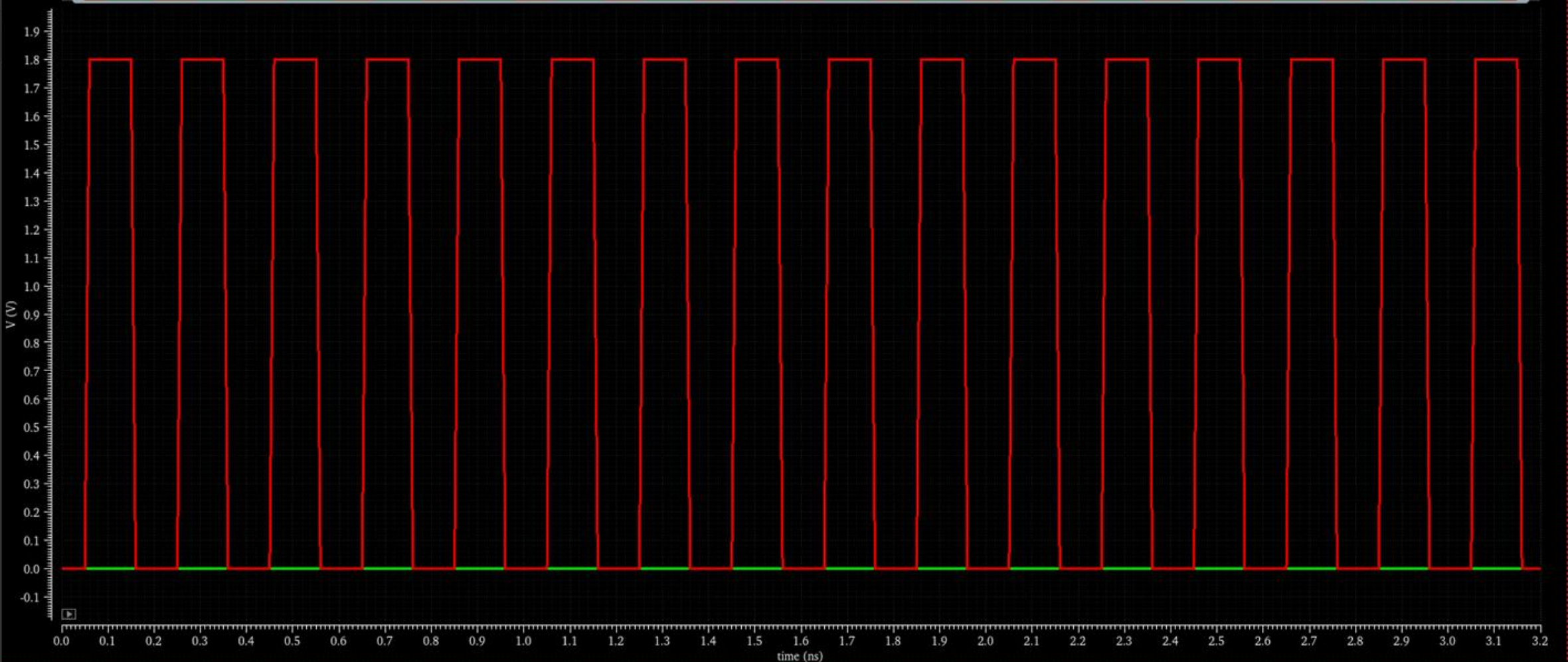
Transient Response

Sat Feb 8 13:08:13 2025 1

Name Vis

/vout

/vin



mouse L:

M:

R:

27(82) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom